

DESIGN SPECIFICATION · ACCOUNTING →
PURCHASING

Three quotes, one memory: purchasing as a system that *learns*.

A sub-module inside HydroNexis-AI Accounting for side-by-side supplier comparison, purchase history pulled from SOS Inventory, and a Nexi-driven RFQ outreach loop that keeps every reply so the next comparison is smarter than the last.

Module · Accounting

View ·

acct_purchasing

Branch · claude/accounting-purchasing-suppliers-5p94sz

Head · 7403d49

Doc rev · 2026-07-27

§ 1

Purpose & scope

What this sub-module is for, and what it deliberately isn't.

Every non-trivial purchase at AteretPaz is meant to be shopped across three suppliers. Today that shopping happens in email threads and paper quotes; nothing carries forward. The Purchasing sub-module is a place to do the comparison once, in one shape, and keep it — so the next time the same product comes up, the answer is already half-written.

It handles: side-by-side supplier comparison (unit / specs / quantity / cost / delivery / other), purchase history per product-supplier pair, a supplier ledger with activity status, and a Nexi-driven outreach flow that dispatches RFQs by email or SMS and logs the reply.

It does not: replace SOS Inventory as the system of record for POs or receipts, replace QuickBooks as the ledger of bills, or approve payments. It is a decision-support and audit layer that reads from those systems and writes intent — quotes requested, quotes received, awards made.

§ 2

Coverage of the handoff outline

Every screen, integration and security item from the initial developer outline — mapped to the current build.

The initial handoff outline lists **16 primary screens**, **5 core integrations** and **6 security requirements**. This section maps each one to what is built, what is partial, and what is roadmap.

Legend: ✓ **DONE** = built and merged on branch · 🟡 **PARTIAL** = data + logic exist, dedicated screen still to build · 📅 **ROADMAP** = not yet built; unlock path noted.

2.1 Primary screens (16)

#	SCREEN	STATUS	WHERE IT LIVES / WHAT TO FINISH
1	Purchasing Mission Control	 PARTIAL	The Purchasing view header carries a live stats banner (comparisons, open, products, suppliers, spend) + integration strip (SOS/QB connection, sync counters). A dedicated “mission control” landing that folds in KPIs, alerts, and drill-through cards is the next step — reuse hnxDashboard render patterns.
2	Purchase Requests	 ROADMAP	A form for a requester to raise a purchase need before it becomes a comparison. Add a new tab “  Requests” storing <code>hydroPro_acct_purch_requests[{id, requester, product, qty, unit, needBy, reason, dept, status:new approved converted}]</code> . “Convert to comparison” button seeds a comparison card.
3	SOS Inventory Check	✓ DONE	Header banner reads <code>window.__hnxSos.getMeta()</code> live; comparison cards include a “SOS item hint” row from <code>_purchSosItemHint(product)</code> — preferred vendor, SKU, last purchase cost, on-hand. Purchase history import from <code>/sos/purchase-orders</code> via <code>_purchSyncFromSOS()</code> .
4	Supplier Discovery	✓ DONE	Nexi outreach modal + <code>_purchGatherAllSuppliers()</code> unions Inventory Master, Accounting Vendors, SOS items’ preferred vendors, past PO vendors, and past-outreach responders. Includes the geo-scoped radius filter (\$10). Web-found suppliers can be added inline and are persisted to Vendors master.
5	Supplier Qualification	 PARTIAL	Suppliers already carry a reliability “score” field on <code>hydroPro_inv_suppliers_v1</code> and Nexi shows a reply-rate chip. A dedicated qualification screen (docs on file, cert expiry, credit check, quality-audit score, blacklist flag) still to build — extend the vendor record with <code>qualification.{status, certs[], credit, docs[], nextReviewOn}</code> .
6	RFQ Builder	✓ DONE	Editable RFQ template with variable substitution (<code>{{product}}</code> <code>{{qty}}</code> <code>{{unit}}</code> <code>{{specs}}</code> <code>{{needBy}}</code> <code>{{supplierContact}}</code> <code>{{requester}}</code> <code>{{companyName}}</code>) in the outreach modal. “Save as default template” persists it in <code>hydroPro_acct_purch_rfq_template</code> .
7	Email / SMS Communication Center	✓ DONE	RFQ dispatch via EmailJS auto-send (silent) → <code>mailto:</code> fallback, and SMS via device <code>sms:</code> URL scheme (body ≤ 380 chars). Every dispatch appended to <code>hydroPro_acct_purch_outreach</code> . The  Nexi RFQ Log tab is the “center” — filter, preview body, mark responses.
8	Quotation Management	✓ DONE	On each  RFQ Log row: response dropdown (pending / quoted / declined / no-response), quoted unit cost, lead days, notes. Marking “quoted” with a unit price auto-

#	SCREEN	STATUS	WHERE IT LIVES / WHAT TO FINISH
			appends a soft quote-record to Purchase History (<code>source: "rfq-quote"</code>) so it surfaces in comparisons.
9	Quote Comparison	✓ DONE	Comparisons tab — up to 3 suppliers per product, side-by-side matrix (unit / specs / qty / unit cost / total / delivery / payment terms / warranty / notes), cheapest badge auto-computed, per-supplier NEW/EXISTING/HAS-HISTORY tagging, per-product history sub-row.
10	Negotiation Center	● PARTIAL	Replies come back into the RFQ Log where the user can update the quote. A dedicated thread view (versioned quotes, counter-offer template, expiry date, revisions log) is roadmap — extend outreach records with <code>revisions[]: [{ts, side, unitCost, deliveryDays, terms, notes}]</code> .
11	Approval Workflow	☐ ROADMAP	Awards are direct today. Add <code>approvalRules{smallLimit, mediumLimit, largeLimit, approverBySize}</code> on the base config; gate <code>_purchAward()</code> to insert an <code>approvalRequest</code> record when total $\geq$ threshold; render a “🔒 Pending approvals” tray for the approver(s).
12	Purchase Order	● PARTIAL	PO data is READ from SOS via <code>_purchSyncFromSOS()</code> and rendered as history. Writing a NEW PO back into SOS is roadmap — add a POST route on the hnx-sync worker (<code>/sos/purchase-orders POST</code>) and hook <code>_purchCreatePO(compId)</code> on award.
13	Receiving	● PARTIAL	Each SOS-imported history row already carries <code>qty + receivedQty</code> ; a “partial” status is preserved when the PO is not fully received. A dedicated receiving screen (per-PO checklist, defects / short-shipment log, three-way match indicator) still to build.
14	Supplier Evaluation	● PARTIAL	📊 Suppliers Ledger rolls up orders count, total spend, last purchase, activity status (ACTIVE / DORMANT / NEW). Reliability score exists on the base record. A scheduled quarterly-review page (delivery on-time %, defect rate, price stability, response speed) is roadmap.
15	Reports	● PARTIAL	CSV export of purchase history is wired (source, status, ref, receivedQty columns included). A dedicated Reports screen with saved views + charts (spend by supplier, spend by category, price variance vs baseline, reply-rate by supplier) is roadmap.
16	Security & Audit	● PARTIAL	Immutability of SOS-imported rows and of RFQ dispatch data is enforced in <code>_purchDeleteHistory()</code> and <code>_purchOutreachSetResponse()</code> . Every write stamps requester from <code>getCurrentUser()</code> . A dedicated audit trail viewer (who / what / when timeline, filterable) still to build. See §12 for encryption + MFA + role notes.

2.2 Core integrations (5)

INTEGRATION	STATUS	HOW IT WIRES IN
SOS Inventory	✓ DONE	Read via <code>window.__hnxAapi("/sos/purchase-orders")</code> for PO history import; <code>window.__hnxSos.getItems()</code> for the item hint row + preferred vendor fallback; <code>window.__hnxSos.getMeta()</code> for connection state. Write-back (create PO) roadmap.
Email	✓ DONE	Silent send via existing EmailJS integration (<code>_sendViaEmailJs</code>). <code>mailto:</code> fallback when EmailJS not configured. Full body archived in the outreach log.
SMS	✓ DONE	Device <code>sms:</code> URL scheme (opens native composer). Silent auto-send is roadmap — Twilio-backed <code>/nexi/sms</code> worker route.
Accounting (QB)	⦿ PARTIAL	Read <code>hydroPro_acct_qb_v1</code> for connection-state hint only. Bill-line ingest is roadmap — add a Make scenario for Bills-by-Vendor-with-Lines feeding <code>_purchSyncFromQB()</code> . Comparisons + PO awards will eventually push bill drafts back.
AI Recommendation Engine	✓ DONE	Registered as a Nexi domain via <code>registerNexiDomain()</code> . Answers ranked-supplier questions (“who supplies X”, “best price on X”, “suppliers near me”, “supplier reply rate”, “how many RFQs sent”) over live stored data. See §9.

2.3 Security & audit (6)

REQUIREMENT	STATUS	HOW IT IS MET
Role-based access	✓ DONE	The Purchasing sub-tab sits inside Accounting and inherits the module's permission gate via <code>VIEW_TO_MODULE / can()</code> . The Accounting role has full manage on this domain; other roles get view-or-none per the existing PERMS engine.
MFA	📅 ROADMAP	Session / MFA is app-wide (currently handled by the cloud-login worker for staff sign-in). No purchasing-specific MFA prompt exists on award/send — recommended to add a step-up MFA prompt before <code>_purchAward()</code> when order value > large-threshold.
Audit trail	🕒 PARTIAL	Every write carries <code>createdAt / updatedAt / updatedBy</code> where the app already has that pattern. Outreach rows are immutable after dispatch (only response fields editable). Dedicated audit viewer TBD — reuse the existing <code>hydroPro_audit_log</code> renderer.
Encryption	📅 ROADMAP	<code>localStorage</code> is not encrypted at rest. For sensitive supplier data recommend either (a) migrating to the app's existing IndexedDB store with a per-user derived key, or (b) proxying reads through the hnx-sync worker with encrypted server-side storage.
Approval limits	📅 ROADMAP	Same as primary #11. Config lives on the base object (<code>approvalRules</code>), gate lives on <code>_purchAward()</code> .
Supplier isolation	✓ DONE	Suppliers keyed by lowercased name; every collection filters by supplier. No supplier can see another's quote — the outreach log is per-supplier per-RFQ, dispatched one message per supplier via EmailJS / mailto / sms; no bcc / mass-send is ever used.

§ 3

Where it lives

Navigation, view registration, discovery.

- › **Primary view** · registered as `acct_purchasing`, dispatched by the app-wide `switchView()` the same way `acct_vendors`, `acct_customers`, `acct_invoices` and `acct_tax` are. Mounted lazily inside `#appMain`.
- › **Module attribution** · added to the `VIEW_TO_MODULE` map with value `accounting`, so the module header banner and permission checks resolve correctly.
- › **Deep-link from Accountant-AI overlay** · a new `purchasing` entry in `ACCT_TABS` shows a summary panel and a “Open Purchasing workspace” button that closes the overlay and calls `switchView('acct_purchasing')`.
- › **Direct URL / console** · `switchView('acct_purchasing')` from anywhere.

§ 4

Four tabs

One header, four working surfaces.

Header banner shows totals: comparisons, open comparisons, products tracked, suppliers used, total logged spend. Underneath, an integration strip: SOS connection state, QB cache state, history-row counts (SOS-imported vs manual), last-sync timestamp, and inline sync controls. Below the header sits the tab strip and the primary action  NEXI · FIND & CONTACT SUPPLIERS .

TAB	WHAT IT DOES	WRITES TO
 Comparisons	Add a product to shop for; capture up to three supplier quotes side by side; expand a card to see the matrix, the cheapest badge, per-supplier history, and a SOS item hint. Actions: award a winner, close, delete, log a purchase from a chosen quote, or ask Nexi for more quotes.	hydroPro_acct_purch_comparisons
 Purchase History	Flat log of every purchase — product, supplier, unit, qty, unit cost, total, PO ref, notes. Filterable search, CSV export. Rows imported from SOS show a SOS badge and cannot be manually deleted.	hydroPro_acct_purch_history
 Suppliers Ledger	Roll-up per supplier: orders count, distinct products, total qty, total spend, last purchase date, activity status (ACTIVE / DORMANT / NEW), recent products chips.	read-only rollup

TAB	WHAT IT DOES	WRITES TO
 Nexi RFQ Log	Every RFQ Nexi dispatched: sent-at, product, supplier, channel, response status, quoted unit cost, lead days, notes. Marking a row <i>quoted</i> auto-appends a soft quote-record to Purchase History so it appears in future comparisons.	hydroPro_acct_purch_outreach

§ 5

Data model

Every persistent shape, in one place.

Everything is stored in `localStorage` under the `hydroPro_*` prefix used across the app. Nothing is written to a backend without going through the existing hnx-sync worker (SOS pulls only).

KEY	SHAPE	PURPOSE
<code>hydroPro_acct_purch_comparisons</code>	array	Saved 3-supplier comparison sheets.
<code>hydroPro_acct_purch_history</code>	array	Merged purchase log — manual entries, SOS-imported PO lines, and RFQ-quote softlines.
<code>hydroPro_acct_purch_outreach</code>	array	RFQ send log with response tracking — the memory Nexi learns from.
<code>hydroPro_acct_purch_sync_meta</code>	object	SOS import status: <code>{status, lastSync, poCount, added, updated}</code> .
<code>hydroPro_acct_purch_rfq_template</code>	string	User-editable RFQ message template with variables.
<code>hydroPro_acct_vendors</code>	array	Existing Accounting Vendors list — Nexi reads it for contact info and writes web-found candidates back.
<code>hydroPro_inv_suppliers_v1</code>	array	Existing Inventory Master suppliers — richer contact data + reliability score.
<code>hydroPro_sos_items_v1</code>	array (IDB)	SOS items cache — read for preferred vendor and last purchase cost per SKU.
<code>hydroPro_emailjs_config_v1</code>	object	Existing EmailJS credentials — reused for silent RFQ auto-send.

Comparison record

```

{
  id: "pc_lqm7...",
  product: "Rockwool cube 100×100",
  category: "Growing media",
  needBy: "2026-08-14",
  requester: "EB",
  status: "open" | "awarded" | "closed",
  chosenSupplier: 0 | 1 | 2 | null,
  createdAt: 1721001600000,
  awardedAt: ...,
  suppliers: [           // up to 3 slots
    { name, unit, qty, specs, unitCost, deliveryDays, payTerms, warranty, notes }
  ]
}

```

Purchase-history record · a single row per PO line or per manual entry.

```

{
  id: "ph_sos_2026-21-P00528-01_0",
  source: "sos" | "manual" | "rfq-quote",
  poNumber: "2026-21-P00528-01", // sos only
  lineIdx: 0,                    // sos only – dedup key
  date: "2026-05-28",
  product: "Business Card 2×3.5",
  supplier: "Micropoint Graphics",
  sku: "",
  unit: "pc",
  qty: 700,
  receivedQty: 0,
  unitCost: 6.75,
  totalAmount: 4725,
  ref: "2026-21-P00528-01",
  status: "open" | "partial" | "closed",
  notes: "From SOS PO · open",
  importedAt: 1721001600000
}

```

Outreach record · one row per RFQ dispatched (per channel per supplier).

```
{
  id: "rfq_lqzmz..._0",
  sentAt: 1721001600000,
  product, qty, unit, specs, needBy, requester,
  supplier, supplierEmail, supplierPhone,
  subject: "RFQ · Rockwool cube 100×100",
  message: "Hello ...",           // fully-rendered body
  channel: "email" | "sms",
  delivery: "emailjs" | "mailto" | "sms-app" | "error",
  status: "sent" | "opened" | "error",    // dispatch result
  responseStatus: "pending" | "quoted" | "declined" | "no-response",
  responseAt: ...,
  quotedUnitCost: ...,
  quotedDeliveryDays: ...,
  responseNotes: "..."
}
```

§ 6

Comparison logic

How the matrix picks the cheapest and reads the history.

Cheapest badge · for each supplier column, `total = qty × unitCost`. Only columns with a positive total are eligible; the lowest wins the  **CHEAPEST** tag. Ties resolve to first-filled.

Award vs cheapest · the buyer's choice can differ from the cheapest — the column with the buyer's award gets a green top-border and a  **CHOSEN** action; the cheapest tag stays independently. When a comparison has no explicit award yet, the cheapest is used as the visual default.

NEW / EXISTING supplier badge · a supplier is *EXISTING* if their name appears in the merged history OR in the Vendors master (case-insensitive, trimmed). Otherwise *NEW*. This is computed at render time — no separate state.

Per-product history sub-row · `_purchLookupHistory(product, supplier)` filters merged history to matching product-supplier pairs and returns `{times, totalQty, totalCost, lastDate, rows[]}`. The card renders the last six line-items so the buyer sees drift in quantity and price without opening another view.

SOS item hint row · `_purchSosItemHint(product)` scans `window.__hnxSos.getItems()` for a name/sku match and returns the item's preferred vendor, SKU, last purchase cost and on-hand. Rendered as a single blue-tinted row between the criteria table and the history rows. If the supplier under review matches SOS's preferred vendor, an extra note surfaces on that column's history cell.

§ 7

Purchase history sources

Three streams merged into one log; every row is source-tagged.

SOS Inventory · `source: "sos"`

`_purchSyncFromSOS()` calls `/sos/purchase-orders` through the existing hnx-sync Cloudflare worker.

Every PO line becomes one history row. Dedup key is `poNumber + lineIdx`, so re-syncs update in place instead of duplicating.

Trigger: manual (button in header banner). Auto-import on first empty load can be added later.

Manual entry · `source: "manual"`

The History tab has an add form: product, supplier, date, unit, qty, unit cost, PO/ref, notes. Also written by “+ Log purchase” on a comparison card, using the chosen supplier’s quote.

Fully editable and deletable.

RFQ quotes · `source: "rfq-quote"`

When an outreach row is marked *quoted* with a unit price, a soft row is auto-appended: `qty:0, unitCost:<quote>`. It surfaces the price to future comparisons without pretending a purchase happened.

Suppresses double-counting because `qty*unitCost=0`.

QuickBooks · deferred

QB currently caches only trial-balance and aging in `hydroPro_acct_qb_v1`, not per-line vendor-item bills. The banner reads its connection state as a hint.

A `_purchSyncFromQB()` can be added in parallel, using the same `source:` dedup pattern; the rest of the UI needs no change.

GUARANTEE

Once `_purchSyncFromSOS()` has run, every card’s per-supplier “n× before · P× total · last DATE” row is real. The buyer never types history — they type comparisons.

§ 8

Nexi outreach flow

From “find suppliers” to logged RFQ, in five steps.

Entry points: the header primary button, or “📄 Ask Nexi for more quotes” on any comparison card. Both open the outreach modal, prefilled with product / qty / unit / need-by from the card if launched from context.

Step 1

Gather candidates

`_purchGatherAllSuppliers(product)` unions Inventory Master, Accounting Vendors, SOS items’ preferred vendors, past PO vendors, and past-outreach responders.

Step 2

Rank & tag

Product-history match > has email > has phone > reliability score > alpha. Each row shows source chips, `EXISTING` / `NEW` / `HAS HISTORY`, and past reply-rate.

Step 3

User picks

Manual checkboxes, or one-click “top 3” / “existing”. A separate form lets the user add a supplier they found on the web — saved to Vendors master so Nexi remembers.

Step 4

Compose

Editable RFQ template with variable substitution: `{{product}}` `{{qty}}` `{{unit}}` `{{specs}}` `{{needBy}}` `{{supplierContact}}` `{{requester}}` `{{companyName}}`.

Step 5

Send & log

Email via existing EmailJS auto-send (silent) or `mailto:`. SMS via device `sms:` URL scheme with body ≤ 380 chars. Every dispatch appends one row to the outreach log with full body archived.

Channel semantics

CHANNEL	DELIVERY MODE	BEHAVIOR
 EMAIL	emailjs	Silent send via existing EmailJS integration. Status "sent". Requires the shared EmailJS config already used for report auto-send.
	mailto	Fallback when EmailJS is not configured. Opens the device mail client with recipient, subject, and body pre-filled. Status "opened" — the app can't verify actual send.
 SMS	sms-app	Opens the device SMS composer via <code>sms:</code> URL scheme. Body auto-shortened to 380 chars with an ellipsis. Status "opened".
	error	Dispatch threw. Full error stored in the row's <code>error</code> field.

§ 9

Nexi intelligence

The learning loop that makes tomorrow’s comparison start with an answer.

The purchasing sub-module registers a Nexi domain via `registerNexiDomain()`, using the app-wide domain registry pattern (Irrigation, Logistics, etc. all register the same way). The registration exposes the three collections and provides a custom `answer(qLower)` function.

ASK NEXI	WHAT SHE READS	WHAT SHE RETURNS
<i>“who supplies rockwool?”</i>	All supplier sources + history	Ranked list with contact info + past-order counts
<i>“best price on rockwool cubes”</i>	Purchase history (incl. RFQ quotes)	Cheapest supplier + next two, with reply-rate
<i>“supplier reply rate”</i>	Outreach log	Per-supplier quoted/sent scoreboard
<i>“how many RFQs have I sent”</i>	Outreach log	Totals + open comparisons + records

THE COMPOUNDING EFFECT

Every outreach is logged. Every quote reply is logged. Every awarded winner is logged. On the third comparison for the same product, Nexi’s ranking already knows: who quoted, who replied within a day, who was cheapest, who won last time. Manual entry is only ever for what hasn’t happened yet.

§ 10

Permissions & audit

Who sees, who edits, what's recorded.

- › The Accounting role has full `manageHR` and payroll access and read-only access to operational modules — the Purchasing sub-tab sits within Accounting and inherits the module's permission gate.
- › Each outreach row records `requester` from the current user profile (`getCurrentUser()`). Once an RFQ has been dispatched, its body and dispatch metadata are immutable; only the response fields (`responseStatus`, `quotedUnitCost`, `quotedDeliveryDays`, `responseNotes`) are editable.
- › SOS-imported history rows are protected from manual deletion — `_purchDeleteHistory()` explicitly refuses them and tells the user to delete the PO in SOS or use "Clear SOS-imported rows".
- › CSV export writes `source`, `status`, `ref`, and `receivedQty` alongside the numeric fields, so an offline audit can trace every row to its origin.

§ 11

Wiring points

Exactly which globals the sub-module touches.

GLOBAL / HOOK	READ?	WRITTEN?	PURPOSE
<code>switchView(v)</code>	hooked	—	Registers <code>acct_purchasing</code> in the dispatcher.
<code>VIEW_TO_MODULE</code>	—	merge	Adds <code>acct_purchasing: "accounting"</code> .
<code>ACCT_TABS</code>	—	splice	Adds the <code>purchasing</code> sub-tab to the Accountant-AI overlay dropdown.
<code>window.__hnxApi(...)</code>	call	—	Fetches <code>/sos/purchase-orders</code> for history import.
<code>window.__hnxSos.getItems()</code>	call	—	Reads cached SOS items for the item hint row.
<code>window.__hnxSos.getMeta()</code>	call	—	SOS connection state for the banner.
<code>_sendViaEmailJs(to, subj, body)</code>	call	—	Silent RFQ auto-send via the existing EmailJS integration.
<code>registerNexiDomain({...})</code>	call	—	Exposes the purchasing intelligence to Nexi's chat/brain.
<code>showToast(msg, tone)</code>	call	—	User feedback on sync / send / save.
<code>getCurrentUser()</code>	call	—	Requester name in the outreach signature.

§ 12

Limits & roadmap

What today's build deliberately doesn't do — and what unlocks each capability.

NOT YET	WHY	WHAT UNLOCKS IT
Automated web supplier search	A browser HTML app can't crawl external sites (CORS, no server).	Add <code>/nexi/supplier-search?q=</code> to the hnx-sync worker; the modal already has a slot for it.
Silent SMS auto-send	Browsers have no direct SMS API; <code>sms</code> : hands off to the device app.	Add a Twilio-backed <code>/nexi/sms</code> worker route; swap the <code>sms</code> : handoff for a POST.
QB bill-line import	Current QB cache is TB/aging only.	Add a Make scenario for <i>Bills by Vendor with Lines</i> ; feed <code>_purchSyncFromQB()</code> .
Automatic reply detection	Response state is human-set today.	An inbound-email webhook (Zapier/Make on your quotes inbox) → <code>/nexi/rfq-reply?rfqId=</code> → auto-flip <code>responseStatus</code> .
Multi-currency & VAT split	Amounts are single-currency, VAT-inclusive.	Add <code>currency</code> , <code>vatRate</code> , <code>vatMode</code> to history rows and quote fields; render alt-currency next to totals.
Approval workflow > PX	Awards are direct today.	Add a threshold rule + approver assignment on the comparison card, gated by role.
Purchase Order creation back into SOS	Only pulls from SOS, doesn't push.	SOS <code>/sos/purchase-orders</code> POST via a new worker route on award.

Nothing on this list breaks the current model. Each item extends a stored field, adds one worker route, or plugs a new record source into the same merged history log.

HydroNexis-AI · Accounting · Purchasing sub-module

Design revision 2026-07-27 · branch `claude/accounting-purchasing-suppliers-5p94sz`

DESIGN SPEC · PURCHASING · FEATURE
II · GEO-SCOPED SUPPLIER SEARCH

Small orders stay local. Big orders *travel farther.*

Nexi filters supplier suggestions by distance from a configurable base, and the search radius scales up with the estimated order value. The bigger the ₱, the wider she looks — because a large order justifies the transport a small one doesn't.

Module · Accounting → Purchasing

Feature · geo-scoped RFQ search

Branch · claude/accounting-purchasing-suppliers-5p94sz

Head · 4fac878

Doc rev · 2026-07-27

§ 1

The rule, in one sentence

Value determines geography.

A ₱4,000 pack of consumables shouldn't be sourced from another country. A ₱1.2M capex order shouldn't be limited to the barangay. The system encodes that judgement so Nexi doesn't need to be reminded, and every buyer follows the same rule.

25 km
100 km
500 km
2 000 km

The rings are the tiers. The base pin is your farm. Each ring is the radius unlocked by a higher order-value bucket. Nexi searches inside the smallest ring the order qualifies for, and widens only if it can't find enough candidates.

Rings are illustrative, not to scale — a real 2 000 km ring wouldn't fit next to a 25 km ring on the same page.

§ 2

Default tier ladder

Sensible starting values — every one editable.

UP TO ₱	RADIUS	MEANING · WHEN IT FIRES
10,000	25 km	Small · local only. Consumables, replacements, urgent top-ups. Freight would dwarf the order.
50,000	100 km	Regular · metro. Weekly resupply, mid-size components. Same-day / next-day trucking still economic.
200,000	500 km	Mid · regional. Bulk media, quarterly packaging runs. Willing to wait a few days for better price.
1,000,000	2,000 km	Large · national. Equipment, project-scale inputs. Whole-country search opens meaningful competition.
∞	∞	Very large · worldwide. Capex, imports, one-off large-ticket. International freight is a rounding error against the deal.

Ladder is edited in the settings modal: 📍 Location & radius. Rows can be added, removed, relabelled, and re-priced; the top row is always the ∞ catch-all. Ordering is by *maxValue* — the first row whose ceiling is ≥ the order value wins.

§ 3

How the value is inferred

The buyer types intent, not arithmetic.

The outreach modal exposes an `Est. value ₱` field. It's not required — Nexi guesses:

- › **From the SOS hint** · if the product matches a SOS item with a `purchaseCost`, estimated value = `qty × lastCost`. This is the primary path once SOS is connected.
- › **From past history** · if any past PO line has the same product name, the median past unit cost × current qty is used as a fallback (roadmap).
- › **Buyer override** · typing a number in the field wins over any guess. The Override km field wins over the tier the value picks. Rules serve the buyer, not the other way around.

§ 4

Auto-widen safety

Nexi never returns zero suppliers because of the geography rule.

Every tier has a floor: `minSuppliersForRadius` (default **3**). If the requested radius surfaces fewer than that many contactable suppliers, Nexi automatically promotes the search to the next larger tier and re-runs.

BEHAVIOUR ON WIDEN

The status strip in the modal flags the widen explicitly — “*auto-widened · fewer than min in the requested tier*” — so the buyer sees when the rule bent and can adjust the tier or the min-count without hunting.

The floor is per-tier by design; the config lives on the base-location object so it moves with the site.

§ 5

Distance computation

Great-circle, not driving distance.

`_purchDistanceKm(lat1, lng1, lat2, lng2)` uses the Haversine formula. It's the right choice for a geography filter: cheap, deterministic, no external service, correct within a fraction of a percent at country-scale distances.

It is *not* driving distance. For a “how far is the truck actually going” number, a future step could call a routing API — but for a filter that decides whether to invite a supplier to quote, great-circle is more than accurate enough, and one straight-line rule is easier to explain than a road network that changes.

§ 6

Base location

One pin. Editable. Optional browser geolocation.

Set once from the header button  Location & radius . The modal captures a display name, an address, latitude and longitude — and offers a  USE MY CURRENT LOCATION button that reads `navigator.geolocation` and fills the coords in place.

Default until edited: Manila metro at `14.5995, 120.9842` . That's a sensible pin for AteretPaz; other sites edit it once and forget it.

```
// hydroPro_acct_purch_base_location
{
  name: "AteretPaz farm",
  address: "",
  lat: 14.5995,
  lng: 120.9842,
  tiers: [
    { maxValue: 10000, radiusKm: 25, label: "Small · local only" },
    { maxValue: 50000, radiusKm: 100, label: "Regular · metro" },
    { maxValue: 200000, radiusKm: 500, label: "Mid · regional" },
    { maxValue: 1000000, radiusKm: 2000, label: "Large · national" },
    { maxValue: Infinity, radiusKm: null, label: "Very large · worldwide" }
  ],
  minSuppliersForRadius: 3,
  includeUnknown: true
}
```

§ 7

Supplier coordinates

Per-supplier lat / lng, with graceful fallback.

A dedicated key stores explicit coordinates per supplier — keyed by lowercased name so a rename in one place doesn't sever the link.

```
// hydroPro_acct_purch_supplier_locations
{
  "micropoint graphics": {
    lat: 14.6532,
    lng: 121.0498,
    address: "Quezon City",
    city: "Quezon City",
    country: "PH"
  },
  ...
}
```

Fallback path. If a supplier has no explicit coord record,

`_purchGetSupplierLocation()` reaches into the existing `hydroPro_inv_suppliers_v1` and `hydroPro_acct_vendors` stores and returns any address / city / country found there — so the row can still show something even if the distance chip has to say “no location · add”.

Adding coords in-flow. Every row in the outreach modal shows a  chip.

Rows without coordinates get a clickable `no location · add` chip that prompts for lat / lng inline; the answer is written straight into the map and the modal re-renders. No mode-switching to a settings page.

ROADMAP

Coord entry is manual today because the app can't geocode from the browser (no server-side routing service). A future `/nexi/geocode?address=` route on the hnx-sync worker would let Nexi auto-fill coords from any address on the vendor record.

§ 8

The outreach modal, changed

Two new controls, one new column, one status strip.

A
Est. value ₱ Auto-guessed from qty × SOS last-cost. Editable. Feeds the tier lookup.
B
Override km Any positive number wins over the tier. Blank means “use the tier”.
C
Status strip 📍 Radius X km · tier Y · in-range vs outside count · widen flag · base name + coords.
D
📍 chip / row Every supplier row shows distance. In-range rows are full opacity, outside rows are dimmed.
E
Selectors Top 3 · Existing · All-in-range — every bulk selector respects the in-range flag.

Nothing is hidden by the filter — out-of-range rows are still visible, just dimmed and unchecked by the bulk selectors, so the buyer can override on a specific supplier without editing the tier ladder.

Nexi can be asked directly

Same rule, natural-language surface.

ASK NEXI	WHAT SHE RETURNS
<i>“suppliers near me”</i>	Ranked list by distance from the base, top 5, with contact info and past-order counts.
<i>“local suppliers for rockwool”</i>	Same, filtered to the product word. Falls back to closest few if no one is inside the default radius.
<i>“suppliers within 50 km”</i>	Parses the number, honours the override, returns everyone inside that radius.
<i>“where is my base”</i>	Reads the base-location record — name and coords.

New keywords: radius , nearby , near me , local , distance , km , kilometer , area . They’re appended to the domain’s keyword array so the registry routes those questions to the geo answerer before falling through to the price / supplier answerers.

§ 10

Storage summary

Two keys added, everything else re-used.

KEY	PURPOSE
hydroPro_acct_purch_base_location	Base name + coords + tier ladder + auto-widen floor + unknown-inclusion flag.
hydroPro_acct_purch_supplier_locations	Map of supplier-lower → {lat, lng, address, city, country}.
hydroPro_inv_suppliers_v1	Read only for fallback address / city / country when a supplier has no explicit coords.
hydroPro_acct_vendors	Same fallback role as above.

`Infinity` can't survive a JSON round-trip, so on save it's written as `null` in the top tier's `maxValue`; on load the config-loader restores it. Any consumer that reads the config through `_purchGetBase()` never sees the sentinel.

Not doing yet

Extension points, called out.

DEFERRED	WHAT UNLOCKS IT
Geocode a plain address	<code>/nexi/geocode?address=</code> on the hnx-sync worker. Would auto-fill supplier coords from stored addresses on Vendors master.
Driving-distance / actual travel time	OSRM or Google Distance Matrix through the worker. Only worth adding if lead-time promises depend on it — otherwise great-circle is enough.
Freight-cost aware radius	Multiply <code>qty × unitCost</code> by a freight-per-km rate before the tier lookup, so the tier reflects <i>delivered</i> cost, not just <i>ordered</i> cost.
Per-category tier overrides	Attach a <code>categoryOverrides</code> map on the base — e.g. “perishables never go past 200 km, regardless of value”.
Multi-site (per-farm) bases	Promote the base record to an array keyed by site; the outreach modal picks the site the buyer selects.

None of the above breaks the current model. Each extension either adds a stored field, adds a worker route, or plugs a new signal into the same tier lookup.

HYDRONEXIS - AI · ACCOUNTING MODULE

PURCHASING SUB-MODULE

Developer specification pack

Complete screen index + 13 developer-ready screen specifications

CONTENTS	Screen Index (91 screens across 18 sections) + Screens 001 → 013 in full technical detail
REPOSITORY	eyalbenari99-star/hydropro
BRANCH	claude/accounting-purchasing-suppliers-5p94sz
STATUS	Ready for Development
VERSION	1.0
DOCUMENT DATE	2026-07-28

Part I — Complete Screen Index

Master navigation for the Purchasing sub-module. Every screen has a unique numeric identifier (e.g. 1.5 for Pending Approval) that will be used as the router key, the URL segment, and the primary reference in all subsequent deliverables.

Total screens: **114** across 18 sections (0–17).

0. Purchasing Home

ID	SCREEN
0.1	Purchasing Mission Control
0.2	Purchasing Calendar
0.3	My Tasks
0.4	Notifications
0.5	AI Assistant (Nexi)

1. Purchase Requests

ID	SCREEN
1.1	New Purchase Request
1.2	Purchase Request Details
1.3	Draft Requests
1.4	Submitted Requests
1.5	Pending Approval
1.6	Approved Requests
1.7	Rejected Requests
1.8	Cancelled Requests
1.9	Archived Requests

2. AI Validation

ID	SCREEN
2.1	AI Review
2.2	Duplicate Detection
2.3	Budget Verification
2.4	Consumption Analysis
2.5	Suggested Alternatives
2.6	Risk Analysis

3. SOS Inventory Integration

ID	SCREEN
3.1	Current Stock
3.2	Available Stock
3.3	Incoming Deliveries
3.4	Reserved Inventory
3.5	Warehouse Transfer Suggestions
3.6	Internal Availability
3.7	Previous Purchase History

4. Technical Specifications

ID	SCREEN
4.1	Product Specification
4.2	Brand Requirements
4.3	Drawing Viewer
4.4	Attachments
4.5	Technical Approval
4.6	Revision History

5. Supplier Management

ID	SCREEN
5.1	Supplier Directory
5.2	Supplier Profile
5.3	Supplier Qualification
5.4	Supplier Categories
5.5	Supplier Performance
5.6	Blacklisted Suppliers
5.7	Preferred Suppliers
5.8	New Supplier Registration

6. Supplier Discovery

ID	SCREEN
6.1	AI Supplier Search
6.2	Local Suppliers
6.3	International Suppliers
6.4	Manufacturer Search
6.5	Distributor Search
6.6	Supplier Comparison

7. RFQ Center

ID	SCREEN
7.1	Create RFQ
7.2	RFQ Wizard
7.3	Send RFQ
7.4	RFQ Tracking
7.5	Supplier Responses
7.6	RFQ Expiration

8. Communication Center

ID	SCREEN
8.1	Email
8.2	SMS
8.3	Communication Timeline
8.4	Attachments
8.5	AI Summary
8.6	Follow-up Scheduler

9. Quotation Management

ID	SCREEN
9.1	Quotation Inbox
9.2	AI Extraction
9.3	Price Comparison
9.4	Technical Comparison
9.5	Commercial Comparison
9.6	Currency Conversion
9.7	Landed Cost

10. Negotiation

ID	SCREEN
10.1	Negotiation Workspace
10.2	Counter Offers
10.3	Price History
10.4	AI Negotiation Suggestions
10.5	Supplier Commitments

11. Evaluation

ID	SCREEN
11.1	Supplier Scorecard
11.2	Risk Score
11.3	Quality History
11.4	Delivery Performance
11.5	Financial Stability
11.6	AI Recommendation

12. Approval Center

ID	SCREEN
12.1	Approval Queue
12.2	Approval Details
12.3	Multi-Level Approval
12.4	Digital Signature
12.5	Audit Log

13. Purchase Orders

ID	SCREEN
13.1	Create PO
13.2	Draft PO
13.3	Approved PO
13.4	Sent PO
13.5	Supplier Confirmation
13.6	Amendment History

14. Receiving

ID	SCREEN
14.1	Expected Deliveries
14.2	Goods Receiving
14.3	Partial Deliveries
14.4	Quality Inspection
14.5	Variance Management
14.6	Close Purchase Order

15. Supplier Evaluation

ID	SCREEN
15.1	Delivery Rating
15.2	Quality Rating
15.3	Service Rating
15.4	Price Rating
15.5	Overall Supplier Score

16. Reports & Analytics

ID	SCREEN
16.1	Spend Analysis
16.2	Supplier Analysis
16.3	Savings Report
16.4	Budget vs Actual
16.5	Procurement KPI Dashboard
16.6	Executive Reports

17. Administration

ID	SCREEN
17.1	Purchasing Settings
17.2	Approval Matrix
17.3	User Roles
17.4	Permissions
17.5	Email Templates
17.6	SMS Templates
17.7	AI Configuration
17.8	Integration Settings
17.9	Security & Audit

Summary counts

SECTION	SCREENS
0. Purchasing Home	5
1. Purchase Requests	9
2. AI Validation	6
3. SOS Inventory Integration	7
4. Technical Specifications	6
5. Supplier Management	8
6. Supplier Discovery	6
7. RFQ Center	6
8. Communication Center	6
9. Quotation Management	7
10. Negotiation	5
11. Evaluation	6
12. Approval Center	5
13. Purchase Orders	6
14. Receiving	6
15. Supplier Evaluation	5
16. Reports & Analytics	6
17. Administration	9
TOTAL	114

SCREEN 001 · ACC-PUR-001

Purchasing Mission Control

Operational control center — 5-second comprehension, no scroll

Purpose

First screen displayed when a user enters Accounting → Purchasing. Operational control center for the Purchasing Department. A trained user must comprehend the entire purchasing operation within 5 seconds. No vertical scroll on a 27" monitor. Critical information only.

Access matrix

ROLE	ACCESS
Purchasing Officer	Full
Purchasing Manager	Full
Accounting Manager	Full
Finance Manager	Read only
CEO	Read only
Warehouse	Limited (deliveries + receiving alerts only)
Department Requester	No access

Header (80 px)

Left-to-right: Company logo · breadcrumb (Accounting → Purchasing → Mission Control) · current user · current date · notifications bell · AI status indicator · global search · profile menu.

Left navigation

- Dashboard
 - Purchase Requests
 - Suppliers
 - RFQs
 - Quotations
 - Negotiations
 - Purchase Orders
 - Receiving
 - Reports
 - Administration
-

Main dashboard layout

12-column responsive grid. Three rows. Nothing below the fold on a 27" monitor at 100% zoom.

Row 1 — KPI cards (5)

#	CARD	DISPLAYS	CLICK TARGET
1	Purchase Requests	New · Pending · Approved · Rejected · Urgent	Opens screen 1.x with matching status filter
2	RFQs	Open · Sent · Waiting · Expired	Opens 7.4 RFQ Tracking, pre-filtered
3	Supplier Responses	Unread · Today · Late · Urgent	Opens 7.5 Supplier Responses
4	Purchase Orders	Draft · Approved · Issued · Receiving · Closed	Opens 13.x
5	Budget Status	Monthly · Committed · Available · Forecast	Opens 16.4 Budget vs Actual

Row 2 — Priority queue + activity timeline

Left panel · AI Priority Queue

Top 20 purchasing tasks requiring attention, ranked by Nexi. Columns: Priority · Request Number · Supplier · Issue · Due Date · AI Recommendation. Row actions: Review · Approve · Assign.

Right panel · Purchasing Activity Timeline

Reverse-chronological. Icons per event type: Created Request · Supplier Replied · PO Approved · Shipment Delayed · Goods Received · Invoice Received · Supplier Warning. Every item click-through to source.

Row 3 — Performance · budget · alerts

PANEL	CONTENTS
Supplier Performance (left)	Top suppliers · Avg delivery · Avg quality · Response time · Risk score · Trend (30/90 days)
Budget Utilization (middle)	Monthly · Department · Project spend · Remaining · Graph (stacked bar per month)
Alerts (right)	Late supplier · Budget exceeded · No supplier found · Duplicate request · Approval waiting · Delivery delay · Critical inventory

Floating AI panel (Nexi)

Always visible, bottom-right. Persistent across navigation inside Purchasing.

- Find supplier
- Compare quotes
- Create RFQ
- Generate PO draft
- Summarize supplier emails
- Risk analysis
- Open chat

Universal search

Single input, keyboard shortcut "/". Searchable entities:

- Item
- Supplier
- Purchase Request
- Purchase Order
- RFQ
- Invoice reference
- SOS item code
- Barcode
- QR

Filters (dashboard-wide)

- Project
- Department
- Supplier
- Category
- Priority
- Status

- Warehouse
- Date range
- Budget code

Filters persist per user session and are echoed in the URL for shareable links.

Live updates

WebSocket subscription. Widgets update in under 2 seconds. Manual refresh not required. If the socket drops, a "reconnecting..." chip appears and the dashboard falls back to a 15-second polling interval.

Notifications

- Approval required
- Supplier reply
- Price change
- Inventory alert
- Delivery delay
- Budget alert
- Security alert

Colour semantics

COLOUR	MEANING
Green	Completed / on-target
Blue	Informational
Orange	Warning · attention required
Red	Critical · immediate action
Grey	Inactive / disabled

Security

- Widget-level RBAC.
- No direct database access from the client.
- Every user action written to AuditLog.
- Session timeout with configurable idle window.
- MFA required for any Approve / Award action.
- All traffic over TLS; WebSocket over WSS.

APIs required

METHOD + PATH	PURPOSE
GET /api/purchasing/dashboard	Row-1 KPI card counts
GET /api/purchasing/alerts	Row-3 alerts panel
GET /api/purchasing/tasks	Row-2 AI Priority Queue
GET /api/purchasing/timeline	Row-2 activity timeline
GET /api/purchasing/kpi	Header + row-3 KPIs
GET /api/suppliers/performance	Row-3 supplier performance
GET /api/budget/status	Row-1 card 5 + row-3 budget
GET /api/sos/inventory	Live inventory hooks for alerts
GET /api/notifications	Header bell
WS /ws/purchasing	Live updates channel

Database tables used

- PurchaseRequests
- PurchaseRequestItems
- Suppliers

- SupplierPerformance
- RFQs
- RFQResponses
- PurchaseOrders
- Receiving
- Budget
- Projects
- Departments
- Users
- Notifications
- AuditLog

KPIs displayed

KPI	DEFINITION
Average Purchase Cycle	Days from PR submission to PO closed
Average Supplier Response	Median time between RFQ sent and first quote received
Savings	(Baseline – awarded price) × qty, monthly rolling
Open RFQs	Count of RFQs whose status is Sent or Waiting
Late Deliveries	PO lines past promised date and not fully received
Budget Utilization	Committed + spent as % of allocated budget, monthly
Supplier Score	Weighted mean of delivery, quality, service, price ratings
Purchase Lead Time	Days from PO issued to first receipt
Inventory Risk	SKUs projected to stock-out within lead time × unit cost
Approval Time	Median hours from PR submitted to first approval action

Acceptance criteria

- Dashboard loads in under 3 seconds on the reference workstation.

- All widgets respect role-based permissions; hidden data never reaches the client.
- Real-time updates occur without page refresh within 2 seconds of the source event.
- All counts reconcile exactly with the underlying transaction data.
- Every alert links directly to the transaction that raised it.
- Every user action is written to AuditLog (actor, timestamp, before/after, IP, device).
- Responsive from 1366×768 to 4K; no horizontal scroll at any size.
- MFA prompt appears before any Approve / Award action.

SCREEN 002 · ACC-PUR-002

Purchase Request

Entry point for every purchasing workflow

Purpose

Creates every purchase request in HydroNexis. Nothing can be purchased without an approved Purchase Request. Every PR receives a unique, immutable PR Number and is traceable through the workflow to a closed PO.

Access matrix

ROLE	CREATE	EDIT	APPROVE	CANCEL
Employee	Yes	Draft only	No	Draft only
Department Manager	Yes	Yes	Level 1	Yes
Purchasing Officer	Yes	Yes	No	Yes
Purchasing Manager	Yes	Yes	Level 2	Yes
Accounting Manager	Yes	Yes	Financial	Yes
CEO	Yes	Yes	Final	Yes

PR Number

Auto-generated on Save Draft.

PR-YYYY-NNNNNN (example: PR-2026-000458)

Immutable once assigned. Sequence is monotonic per calendar year; resets on 1 January.

Header fields (read-only after creation)

- Purchase Request No.
- Status
- Created By
- Created Date
- Last Modified
- Revision
- Priority
- Workflow Stage

Workflow

#	STAGE	EXIT CONDITION
01	Draft	User clicks Submit and all Submit-Validation checks pass
02	Submitted	AI Validation job completes
03	AI Validation	No blocking issues, or user acknowledges warnings
04	Department Approval	Level 1 approver decision recorded
05	Purchasing Review	Purchasing Officer / Manager assigns handling
06	Inventory Verification	SOS reconfirms need after pending transfers
07	Supplier Sourcing	Suppliers selected or Supplier Discovery triggered
08	RFQ	RFQ sent; awaiting responses
09	Quotation Evaluation	At least one qualifying quote received and scored
10	Approval	All required approval levels recorded
11	Purchase Order	PO created and sent
12	Receiving	Goods received against PO
13	Completed	PO closed, three-way match complete

Main form — general information

FIELD	TYPE	RULE	REQUIRED
Request Title	Text	Max 150 chars	Yes
Department	Dropdown	Sourced from Departments	Yes
Project	Dropdown	Sourced from Projects	No
Cost Center	Dropdown	Accounting integration	Yes
Budget Code	Lookup	Accounting Budget	Yes
Required Delivery Date	Date	≥ today; category-constrained	Yes
Priority	Enum	Normal · High · Urgent · Emergency	Yes
Justification	Text area	Min 30 · Max 3000 chars	Yes

Item grid

A single PR may contain multiple items. Every row is independently validated.

COLUMN	RULE
Line No.	Auto 1..N per PR
Item Code	Free text OR lookup into SOS Items · triggers Intelligent Item Recognition
Item Description	Populated from lookup, editable
Category	Inherited from item; overridable
Quantity	Positive number · decimals per unit definition
Unit	From item master
Estimated Unit Price	Populated from last purchase · editable
Estimated Total	Computed: qty × unit price · read-only
Currency	Defaults to company currency · overridable per line
Preferred Brand	Optional; drives supplier suggestion ranking
Alternative Brand Allowed	Boolean; if false, Nexi must not suggest substitutes
Warehouse Destination	Which warehouse receives the goods
Required Date	Per-line override of header Required Delivery Date
Remarks	Free text
Attachment	Per-line file upload

Intelligent item recognition

When an Item Code is entered, Nexi queries every source and returns results inline within 500 ms:

- SOS Inventory (on-hand + reserved)
- Previous purchases of this item
- Supplier database
- Warehouse availability (all sites)
- Historical pricing (12 months, per supplier)
- Open POs for the same item
- Approved brands
- Existing draft/submitted duplicate requests

Attachments

RULE	VALUE
Formats	PDF · DWG · DXF · Excel · Word · Images · Videos · ZIP
Max size per file	500 MB
Virus scan	Every upload scanned before persistence; failed scans quarantined and audit-logged
Retention	Tied to the PR record; default 7 years

Budget validation

Real-time. Every quantity or unit-price change re-runs the check. Checks: Budget Code availability · period remaining · Project budget · Department budget · monthly and annual allocation.

RESULT	BEHAVIOUR
OK	Green tick next to the amount
Insufficient	Yellow warning — submission allowed with manager acknowledgement
Exceeded	Red block — Submit disabled until fixed or override granted

Inventory validation

Calls SOS Inventory on every recomputation: current stock · reserved · incoming · warehouse transfers in flight · alternative items.

RULE

If inventory already exists that satisfies the request, Nexi recommends internal transfer and marks the line as blocked pending acknowledgement.

Duplicate detection

Searches for open or submitted PRs matching all of: same item · same department · same project · same week · same preferred supplier · quantity band ±20%.

Buttons

BUTTON	EFFECT
Save Draft	Persists as Draft; PR Number assigned on first save
Submit	Advances to AI Validation stage
Validate	Runs full validation without submitting
Check Inventory	Forces fresh SOS query
Find Supplier	Opens 6.1 AI Supplier Search seeded
Estimate Cost	Refreshes unit-price estimates
Cancel	Cancels PR (respects Access matrix)
Print	Print-friendly view
Export PDF	PDF with QR link back
Clone Request	Duplicates into new Draft

APIs

METHOD + PATH	PURPOSE
POST /api/purchasing/pr	Create new PR (Draft)
PUT /api/purchasing/pr/{id}	Update fields or line items
GET /api/purchasing/pr/{id}	Fetch a single PR with items + workflow state
POST /api/purchasing/pr/{id}/submit	Advance to AI Validation
POST /api/purchasing/pr/{id}/cancel	Cancel the PR
POST /api/purchasing/pr/{id}/clone	Clone into a new Draft
GET /api/sos/items	Item lookup
GET /api/sos/stock	Current + reserved stock
GET /api/budget/check	Real-time budget validation
GET /api/projects	Project dropdown
GET /api/departments	Department dropdown
GET /api/suppliers/history	Supplier intelligence panel
POST /api/attachments/upload	Multi-part upload
POST /api/ai/pr/validate	Run AI validation

Database tables

- PurchaseRequest
- PurchaseRequestItems
- PurchaseRequestWorkflow
- PurchaseRequestAttachments
- Departments
- Projects
- Budget
- Inventory

- Suppliers
- AuditLog
- Notifications

Acceptance criteria

- PR number is generated automatically and is immutable.
- A single PR supports N line items with independent, per-line validation.
- Budget checks and SOS inventory checks are real-time.
- Duplicate requests are surfaced before submission and require explicit acknowledgement.
- AI recommendations are non-blocking by default; policy flags can make specific ones blocking.
- All changes captured by the audit trail.
- Submission creates the workflow instance and notifies the next approver within 5 seconds.
- A submitted PR cannot be edited except through the versioned Revision mechanism.

SCREEN 003 · ACC-PUR-003

AI Purchasing Analysis

SOS Inventory Intelligence — prevent unnecessary purchasing

Purpose

Immediately after a PR is submitted, Nexi performs a complete procurement analysis before any supplier is contacted. Prevent unnecessary purchasing. First determine whether the company already has the required item, an approved substitute, an incoming delivery, or another internal source.

Processing time

Target: less than 5 seconds. Live progress indicator shows every step.

- Checking SOS Inventory...
- Checking all warehouses...
- Checking open Purchase Orders...
- Checking incoming deliveries...
- Checking approved alternatives...
- Checking historical purchases...
- Checking supplier contracts...
- AI Analysis Complete.

Screen layout

Left panel — Purchase Request summary

Read-only. Displays: PR Number · Requester · Department · Project · Required Date · Priority · Total Estimated Cost · Line Items · Approval Status.

Center panel — AI analysis

Each requested item receives an analysis card. Example:

FIELD	VALUE
Item	2 HP Stainless Water Pump
Inventory Status	Main 0 · GH1 1 · GH5 2 · GH10 0
Total Available	3 units
Required	2 units
Recommendation	Internal Transfer
Status	Purchase Not Required
Savings	₱48,000
Confidence	98%

Alternative item analysis

If the requested item is unavailable, searches: approved equivalents · compatible brands · previous substitutions · engineering approvals · manufacturer compatibility.

REQUESTED	SUGGESTED	COMPATIBILITY
Pump A	Pump B	99%

Incoming deliveries

Checks approved POs. Example: Supplier ABC Industrial · Delivery tomorrow · Qty 10 · Reserved 2 · Available 8 · Recommendation: Delay Purchase.

Existing contracts

Searches: Framework agreements · Annual contracts · Corporate agreements · Negotiated pricing.

FIELD	DISPLAYED
Supplier	Name
Contract Number	Ref
Expiration	Date
Price	Locked rate
Remaining Allocation	Units / value

Warehouse transfer analysis

Instead of buying: can another warehouse supply it? Example: Move GH8 → Main · Transport ₱450 · Purchase cost ₱24,000 · Recommendation: Transfer.

Historical price analysis

- Last Purchase Price
- Average Price
- Highest Price
- Lowest Price
- Current Estimate
- Inflation
- Price Trend
- Price Volatility
- Price Prediction

Example: Average ₱2,450 · Expected market ₱2,320 · Recommendation: Wait 7 days · Estimated savings ₱18,500.

Consumption analysis

- Average Monthly Usage
- Maximum Usage
- Minimum Usage
- Current Consumption
- Inventory Turnover

- Forecast

Example: Current inventory covers 4 months · Recommendation: No purchase required.

Duplicate purchase detection

Searches: all PRs · open RFQs · open POs · recent deliveries. Shows any duplicate, e.g. PR-2026-00481 same department, same item, same quantity. Decision: Merge Requests.

Supplier availability

Without sending an RFQ yet. Displays: approved suppliers · last supplier · rating · avg delivery · lead time · quality · response time · warranty claims.

AI Decision Panel

Displays one final recommendation. Possible results:

- Purchase Immediately
- Transfer Inventory
- Wait for Delivery
- Use Existing Contract
- Delay Purchase
- Use Alternative Item
- Combine Purchase Requests
- Reject Request
- Escalate to Engineering
- Escalate to Finance

Each recommendation includes: Confidence % · Estimated Savings · Estimated Delay · Risk Level · Reasoning.

User actions

- Approve AI Recommendation
- Override Recommendation
- Request Manual Review

- Proceed to Supplier Search
- Back to Purchase Request
- Export Analysis

Override rules

If a user overrides Nexi, they must enter: Reason · Business Justification · Approval (if required). Override is permanently recorded.

Integrations (read-only)

- SOS Inventory
- Warehouse
- Projects
- Accounting
- Supplier Database
- Contracts
- Purchase Orders
- Receiving

Required APIs

METHOD + PATH	PURPOSE
GET /api/sos/stock-all	All-warehouse stock snapshot
GET /api/sos/transfers	In-flight transfers
GET /api/contracts	Active supplier contracts
GET /api/purchase-orders/open	Open POs relevant to the requested items
GET /api/inventory/history	Consumption + turnover history
GET /api/forecast	Consumption forecast
GET /api/price-history	Historical + market price series
GET /api/suppliers	Supplier availability lookup
POST /api/ai/purchase-analysis	Runs the complete analysis, returns the AI decision packet

Database tables

- Inventory
- WarehouseInventory
- InventoryTransfers
- PurchaseOrders
- PurchaseOrderItems
- SupplierContracts
- HistoricalPrices
- ForecastConsumption
- PurchaseRequest
- SupplierPerformance
- AuditLog

KPIs

- Purchases Avoided
- Internal Transfers Recommended
- Savings from Inventory Reuse
- Duplicate Requests Prevented
- Contract Utilization
- Average Inventory Coverage
- AI Recommendation Accuracy

Acceptance criteria

- Analysis completes in under 5 seconds under normal load.
- Every line item is evaluated independently.
- Recommendations include confidence, estimated savings, and rationale.
- Overrides require justification and are fully audited.
- Users can proceed to Supplier Search only after the analysis is complete or an override is approved.
- All inventory, contract, and PO data are read live from integrated systems.

ARCHITECTURE RECOMMENDATION

This screen should be the MANDATORY AI GATE. No RFQ may be issued until this screen completes successfully. Ensures the company does not contact suppliers when stock, transfers, contracts, or incoming deliveries already satisfy the need.

SCREEN 004 · ACC-PUR-004

Supplier Discovery & Intelligence

Discover, evaluate, rank and recommend suppliers before any RFQ is sent

Purpose

Automatically identify the best suppliers for the requested item. Not simply "find suppliers." Nexi intelligently discovers, evaluates, ranks and recommends suppliers before any RFQ is sent.

Trigger

Automatically opens after AI Purchasing Analysis returns PURCHASE REQUIRED.

Objectives

- Local Suppliers
- International Suppliers
- Manufacturers
- Distributors
- Existing Approved Suppliers
- New Potential Suppliers

Search sources

Internal

- HydroNexis Supplier Database
- Previous Purchases
- Approved Vendor List
- Supplier Performance History
- Contracts
- SOS Inventory Supplier Records

External (configurable per company policy)

- Manufacturer websites
- Official distributor databases
- Supplier directories
- Trade association databases
- Approved procurement portals

Search criteria

- Item Description
- Item Code
- Manufacturer
- Brand
- Category
- Country
- Lead Time
- MOQ
- Budget
- Incoterms
- Required Certifications
- Warranty
- Currency

- Language

Search filters

FILTER	OPTIONS
Location	Philippines · Asia · Worldwide
Supplier type	Manufacturer · Distributor · Dealer · Importer · OEM
Business type	Corporation · SME · Government · Preferred Vendor
Risk	Low · Medium · High
Certification	ISO · FDA · GAP · Organic · CE · UL · RoHS · Custom

Results — Supplier Intelligence Card

Each supplier appears as a card. Displays:

- Company Name
- Logo
- Country
- Years in Business
- Products Offered
- Supplier Type
- Rating
- Risk Level
- Preferred Status
- Last Purchase
- Lead Time
- Minimum Order
- Response Time
- Payment Terms
- Warranty
- Delivery Rating

- Quality Rating
- Overall Score

Supplier 360°

Selecting a supplier opens the Supplier Profile. Displays:

- Company Information
- Business Registration
- Tax Registration
- Address
- Website
- Primary / Secondary / Emergency Contact
- Email · SMS · WhatsApp (optional)
- Bank Information (restricted)
- Insurance
- Certifications
- Factory Locations
- Warehouse Locations
- Service Centers

Historical purchasing

- Total Purchases
- Last Purchase Date
- Average Purchase Value
- Average Delivery Time
- Late Deliveries
- Rejected Deliveries
- Warranty Claims
- Price Stability
- Relationship Duration

Supplier performance KPIs

- On-Time Delivery %

- Quality %
- Average Response Time
- Quotation Accuracy
- Invoice Accuracy
- Communication Score
- Complaint Count
- Corrective Actions
- Overall AI Rating

AI Supplier Score — default weighting

CRITERION	WEIGHT
Commercial Competitiveness	20%
Quality History	20%
Delivery Performance	15%
Financial Stability	10%
Response Speed	10%
Pricing Trend	10%
Risk Assessment	10%
Sustainability	5%

Administrators can adjust the weighting.

AI recommendations

- Preferred Supplier
- Lowest Cost
- Fastest Delivery
- Highest Quality
- Lowest Risk

- Best Overall Value
- Existing Contract Available
- Alternative Supplier Recommended

Duplicate detection

Warn if: supplier already responding to another RFQ · suspended · blacklisted · under investigation · contract expired.

Risk engine

- Financial Risk
- Country Risk
- Currency Risk
- Supply Chain Risk
- Delivery Risk
- Quality Risk
- Compliance Risk
- Cybersecurity Risk (for digital vendors)

Each risk is scored and color-coded.

Communication readiness

- Verified Email
- Verified Phone
- Preferred Contact Method
- Business Hours
- Time Zone
- Primary Language
- Average Reply Time
- Last Contact Date

Multi-supplier selection

User can select multiple suppliers for the RFQ. Displays: estimated competition level · expected response probability · coverage of all requested items · recommended number of suppliers.

Actions

- View Profile
 - Compare Suppliers
 - Select for RFQ
 - Add to Preferred Vendors
 - Flag for Review
 - Suspend Supplier
 - Create New Supplier
 - Export Supplier Profile
-

Security

Only Purchasing Managers and Administrators can: approve new suppliers · suspend suppliers · modify supplier ratings · view bank information. Everyone else has role-based restrictions.

APIs

METHOD + PATH	PURPOSE
GET /api/suppliers/search	Cross-source supplier search
GET /api/suppliers/profile/{id}	Supplier 360 profile
GET /api/suppliers/history/{id}	Historical purchasing
GET /api/suppliers/performance/{id}	KPI panel
GET /api/contracts/{supplierId}	Contract lookup
POST /api/suppliers/qualification	Kick off qualification
POST /api/suppliers/select	Select for RFQ

Database tables

- Suppliers
 - SupplierContacts
 - SupplierPerformance
 - SupplierCertificates
 - SupplierContracts
 - SupplierRisk
 - SupplierRatings
 - SupplierCommunications
 - SupplierCategories
 - SupplierBankAccounts (encrypted)
 - AuditLog
-

Acceptance criteria

- Search returns results from all enabled data sources.
- Every supplier receives an AI score and visible explanation.

- Users can compare multiple suppliers side by side.
- Only authorized users can modify supplier records or view sensitive information.
- Selected suppliers flow directly into the RFQ process without re-entry.
- Every supplier action is written to the audit log.

ARCHITECTURE RECOMMENDATION

Build a SUPPLIER KNOWLEDGE GRAPH behind this screen. Instead of storing suppliers as isolated records, Nexi maintains relationships between: suppliers · products · categories · purchase history · projects · warehouses · contracts · performance · quality issues · employees who worked with them. Enables questions like "Who is the best supplier we've ever used for stainless pumps?" and "Which suppliers have the lowest total cost of ownership over 5 years?"

SCREEN 005 · ACC-PUR-005

RFQ Management Center

Prepare, distribute, track, follow up, and evaluate supplier responses

Purpose

Manages the complete quotation process. Instead of manually emailing suppliers, Nexi prepares, distributes, tracks, follows up, and evaluates responses. One RFQ may contain one or many PR items. One RFQ may be sent to one or many suppliers.

Entry conditions

- Purchase Request Approved
 - AI Purchasing Analysis Complete
 - Supplier Selection Complete
-

Header

RFQ Number: RFQ-YYYY-NNNNNN

Status: Draft · Sent · Partially Responded · Completed · Cancelled · Expired

Buyer:

Creation Date:

Closing Date:

Currency:

Revision Number:

Left panel — RFQ summary

- Purchase Request Number
- Department
- Project
- Budget
- Total Estimated Value
- Number of Items
- Number of Suppliers
- Closing Date
- Priority

Timeline

Draft → Approved → Sent → Supplier Opened → Supplier Replied → Clarification → Quotation Complete → Comparison Ready

Center panel — items included

COLUMN	NOTES
Line	Auto per RFQ
Item	Ref to PR item
Description	From item master
Specification	Preview inline
Quantity	Number
Unit	From item master
Required Delivery Date	Per line, may override header
Attachment	Drawings open inline
Comments	Free text

Supplier list

COLUMN	NOTES
Supplier Name	
Country	
Contact Person	
Email	Verified
SMS	Verified
Last Response Time	
Supplier Rating	
Status	see below
Selected	Checkbox

Supplier status values

- Draft
- Queued
- Sent
- Delivered
- Opened
- Downloaded Attachment
- Replied
- Partial Reply
- Completed
- No Response
- Declined

Right panel — Nexi RFQ Assistant

- Increase Supplier Count
- Include Manufacturer
- Split RFQ
- Combine RFQ
- Recommended Closing Date
- Suggested Delivery Terms
- Suggested Payment Terms
- Risk Warnings

RFQ generation

Nexi automatically creates a Professional RFQ PDF containing: technical specifications · commercial requirements · delivery requirements · warranty requirements · company terms · confidentiality statement · submission instructions · attachments. No manual composition required.

Communication methods

- Email
- SMS
- Supplier Portal
- API (future)
- EDI (future)

Buyer chooses one or more.

Email tracking (per supplier)

- Sent
- Delivered
- Opened
- Clicked
- Downloaded Attachment
- Replied
- Bounce
- Failure
- Spam
- Timestamp

SMS tracking (per supplier)

- Sent
- Delivered
- Read
- Reply Received
- Failed
- Retry

Supplier Portal (recommended)

Each supplier receives a secure one-time-token link (no login). Supplier can: Download RFQ · Ask Questions · Upload Quotation · Upload Certificates · Update Delivery · Accept NDA · Submit Response · View Deadline. Link expires automatically after the RFQ closes.

Clarification management

Every supplier question becomes a ticket visible to Purchasing, Engineering, Requester. Answers can be private (only to that supplier) or shared with all participating suppliers to ensure fairness.

AI monitoring

- Which suppliers have not opened the RFQ
- Who opened but did not respond
- Expected response probability
- Recommended reminders
- Suppliers likely to decline
- Supplier workload

Automatic follow-up (configurable)

DAY	ACTION
Day 2	Reminder Email
Day 4	Reminder SMS
Day 6	Buyer Notification
Day 7	AI Recommendation: invite additional supplier

Attachment manager

- Supports PDF · DWG · DXF · Word · Excel · ZIP · Photos · Videos
- Checksum · Virus scan · Version number · Digital signature (optional) per file

RFQ version control

Any modification creates a Revision (number, modified by, reason, affected suppliers). Suppliers automatically receive the updated version. Old versions remain available for audit.

Buyer actions

- Create RFQ
- Preview RFQ
- Edit
- Duplicate
- Save Draft
- Send
- Pause
- Cancel
- Extend Closing Date
- Add Supplier
- Remove Supplier
- Generate PDF
- Export Excel
- Archive

AI risk engine — alerts

- Too Few Suppliers
- Supplier Monopoly

- Budget Too Low
- Specification Too Restrictive
- High Risk Supplier
- Unusually Short Deadline
- Potential Collusion Indicator
- Contract Conflict

Security

Suppliers cannot see: other suppliers · other quotations · internal notes · budgets · AI recommendations · internal comments. Every supplier has isolated access.

APIs

METHOD + PATH	PURPOSE
POST /api/rfq/create	Create RFQ
PUT /api/rfq/update	Update RFQ (creates revision)
POST /api/rfq/send	Dispatch to selected suppliers
GET /api/rfq/status	Aggregate + per-supplier status
GET /api/rfq/suppliers	Suppliers on this RFQ
POST /api/email/send	Email dispatch
POST /api/sms/send	SMS dispatch
POST /api/rfq/portal	Provision one-time-token portal link
POST /api/rfq/reminder	Manual reminder

Database tables

- RFQ

- RFQItems
- RFQSuppliers
- RFQAttachments
- RFQVersions
- RFQCommunications
- RFQQuestions
- RFQResponses
- EmailQueue
- SMSQueue
- AuditLog

KPIs

- Average RFQ Cycle Time
- Supplier Response Rate
- Average Response Time
- No Response Rate
- RFQ Completion Rate
- Average Suppliers per RFQ
- Reminder Effectiveness

Acceptance criteria

- An RFQ cannot be sent without at least one approved supplier.
- Every supplier receives only the information intended for them.
- All communications are tracked with timestamps.
- Revisions are version-controlled and auditable.
- Automated reminders follow configurable schedules.
- Every supplier response is linked back to the originating RFQ and PR.
- The buyer has a complete real-time view of supplier engagement from a single screen.

ENHANCEMENT – DIGITAL PROCUREMENT WORKSPACE

For each RFQ, create a secure workspace centralising: RFQ documents · technical clarifications · email history · SMS history · uploaded quotations · certificates · delivery commitments · negotiation history · AI summaries · approval history. The buyer never searches through Outlook or WhatsApp because every procurement conversation lives with the RFQ itself.

SCREEN 006 · ACC-PUR-006

AI Quotation Analysis & Comparison

Automatic extraction, comparison, TCO calculation, best-supplier recommendation

Purpose

Automatically receive supplier quotations, extract all commercial and technical information, compare every quotation, detect inconsistencies, calculate the true procurement cost, and recommend the best supplier. The Purchasing Officer should never manually build Excel comparison sheets again.

Entry conditions

- RFQ closed or minimum quotation requirement reached
 - At least one supplier quotation received
 - RFQ status = "Ready for Evaluation"
-

Layout

Three-panel interface.

Left panel — RFQ information

- RFQ Number
- Purchase Request
- Project
- Department
- Buyer

- Closing Date
- Number of Suppliers Invited
- Number of Quotations Received
- Currency
- Total Estimated Budget

Center panel — supplier comparison matrix

One row per supplier. Columns:

- Supplier
- Overall AI Score
- Commercial Score
- Technical Score
- Quality Score
- Delivery Score
- Risk Score
- Warranty Score
- Compliance Score
- Total Cost
- Recommendation
- Status

Selecting a supplier expands complete quotation details.

Right panel — Nexi recommendation

- Recommended Supplier
- Second Choice
- Savings
- Risks
- Concerns
- Negotiation Opportunities
- Confidence
- Overall Recommendation

AI quotation extraction

When quotation files arrive, Nexi automatically extracts:

- Supplier Name
- Quotation Number
- Date
- Currency
- Item List
- Unit Prices
- Discounts
- Taxes
- Freight
- Insurance
- Lead Time
- Delivery Terms
- Payment Terms
- Warranty
- Validity
- Incoterms
- Special Conditions
- Minimum Order
- Certificates

No manual encoding.

File types & OCR

- Formats: PDF · Excel · Word · Images · Scanned Documents · ZIP · Email Attachments
- If scanned: run OCR to extract tables, prices, text, signatures, dates, item codes automatically

Technical comparison

FIELD	DISPLAYED
Requested Specification	from PR / RFQ
Supplier Specification	from quotation
Compliance %	computed
Missing Features	
Additional Features	
Engineering Comments	free text
AI Technical Rating	0-100

Commercial comparison

- Unit Price
 - Discount
 - Extended Price
 - Taxes
 - Shipping
 - Insurance
 - Installation
 - Training
 - Support
 - Maintenance
 - Spare Parts
 - Grand Total
-

Landed cost calculation

Real purchasing cost. Formula:

Total Ownership Cost = Purchase + Shipping + Insurance + Import Duties + Brokerage
+ Transport + Installation + Commissioning + Training + Expected Maintenance

This — not just purchase price — drives supplier ranking.

Delivery comparison

Example: Supplier A 14 days · Supplier B 21 days · Supplier C 45 days. Also evaluates historical delivery accuracy, average delay, delivery reliability.

Warranty analysis

- Warranty Duration
- Coverage
- Replacement Time
- On-site Service
- Parts Availability
- Warranty History

Quality history

- Accepted %
- Rejected %
- Returned %
- Corrective Actions
- Complaints
- Quality Trend

Financial analysis

- Price Stability
- Currency Exposure
- Payment Terms
- Credit Limit
- Financial Risk
- Outstanding Issues

AI risk detection

- Unusually Low Price
- Missing Specification
- Hidden Costs
- Incomplete Quote
- Expired Quotation
- Suspicious Discount
- Supplier Dependency
- Collusion Indicators

Comparison modes

- Lowest Price
- Fastest Delivery
- Best Quality
- Lowest Risk
- Best Total Value
- Custom Weighted Score

AI Recommendation Engine

Produces: Executive Summary · Recommended Supplier · Second Choice · Why · Savings · Risks · Negotiation Suggestions · Expected ROI · Confidence Score. No black-box decisions — every recommendation includes a visible explanation.

Buyer actions

- Open Quotation
 - Compare
 - Request Clarification
 - Start Negotiation
 - Reject Supplier
 - Shortlist
 - Recommend Award
 - Export Comparison (PDF / Excel)
-

Visualisations

- Price Distribution
 - Supplier Score Radar
 - Lead Time Comparison
 - Quality Trend
 - Cost Breakdown
 - Delivery Performance
 - Historical Price Trend
-

Security

Quotations cannot be modified. Original supplier documents remain immutable. All extracted values link back to the original document. Only Purchasing Managers and authorized Accounting users can finalize supplier recommendations.

APIs

METHOD + PATH	PURPOSE
POST /api/quotation/upload	Upload supplier quotation
POST /api/quotation/extract	Run OCR + extraction
GET /api/quotation/compare	Compute comparison matrix
GET /api/quotation/technical	Technical comparison payload
GET /api/quotation/commercial	Commercial comparison payload
GET /api/quotation/risk	Risk detection results
POST /api/quotation/recommendation	Produce recommendation packet

Database tables

- Quotation
 - QuotationItems
 - QuotationExtraction
 - QuotationComparison
 - SupplierScores
 - CommercialEvaluation
 - TechnicalEvaluation
 - RiskAssessment
 - Recommendation
 - AuditLog
-

KPIs

- Average Savings Achieved
- Average Evaluation Time
- AI Extraction Accuracy

- Negotiation Opportunity Value
- Average Supplier Score
- Evaluation Completion Time
- Total Cost of Ownership Savings

Acceptance criteria

- Quotations are parsed automatically from supported file types.
- Every extracted value is traceable to the source document.
- Technical and commercial evaluations are displayed side by side.
- Total Cost of Ownership is calculated consistently for every supplier.
- AI recommendations include explanations, confidence scores, and identified risks.
- All recommendation changes and user overrides are fully audited.

ENHANCEMENT — EXPLAINABLE AI

Every recommendation must answer "Why?" with evidence. Example — Recommended Supplier: ABC Industrial. Reasoning: Lowest TCO ₱2,845,000 · 98.4% historical on-time delivery · 0 warranty claims 24 months · 99.2% specification compliance · existing framework agreement reduces cycle 6 days · savings vs next-ranked ₱187,450. Buyer can expand each point to see the underlying data and calculations.

SCREEN 007 · ACC-PUR-007

AI Negotiation Center

Nexi as the buyer's intelligent negotiation assistant

Purpose

After quotations are evaluated, Purchasing begins negotiations with shortlisted suppliers. Nexi becomes the buyer's intelligent negotiation assistant — recommending negotiation strategies, identifying opportunities, tracking concessions, and maintaining a complete negotiation history.

Objectives

- Reduce purchase cost
 - Improve payment terms
 - Improve delivery time
 - Increase warranty
 - Reduce risk
 - Maintain supplier relationship
 - Record every negotiation
-

Entry conditions

- RFQ Evaluation Completed
- Supplier Shortlist Approved
- Negotiation Required

Layout — three panels

Left panel — supplier list

- Supplier
- Current Rank
- AI Score
- Current Offer
- Negotiation Status
- Last Contact
- Response SLA
- Risk Indicator

Color codes:  Active ·  Waiting Response ·  Overdue ·  Closed

Center panel — negotiation workspace

Every communication is displayed chronologically. Channels: Email · SMS · Supplier Portal · Internal Notes. Each message contains: Date & Time · Sender · Recipient · Subject · Attachments · AI Summary · Read Status. Nothing is deleted.

Right panel — Nexi negotiation assistant

FIELD	EXAMPLE
Current Supplier Price	₱1,250
Historical Average	₱1,180
Best Competitor	₱1,165
Recommended Counter Offer	₱1,170
Confidence	93%
Expected Acceptance	78%
Estimated Savings	₱184,000

Negotiation categories

- Price
- Lead Time
- Warranty
- Payment Terms
- Delivery Schedule
- Packaging
- Training
- Installation
- Technical Support
- Spare Parts
- After-Sales Service
- Penalty Clauses
- Service-Level Agreement
- Incoterms

AI negotiation analysis

- Price Reduction Potential
- Delivery Improvements
- Warranty Gaps
- Commercial Risks
- Missing Conditions
- Hidden Costs
- Supplier Weaknesses
- Competitive Advantages
- Historical Negotiation Success

Concession tracker

CATEGORY	BUYER	SUPPLIER
Price	Requested 8%	Approved 5%
Delivery	30 Days	21 Days
Warranty	12 Months	24 Months
Payment	30 Days	45 Days

Offer comparison

Revision 1 → Revision 2 → Revision 3 → Final Offer

Differences are highlighted automatically between revisions.

AI opportunity detection

- Supplier reduced price for another project previously
- Delivery can be improved based on historical performance
- Payment terms better than requested
- Free installation available
- Warranty extension likely
- Annual contract discount possible

Negotiation playbook (configurable)

TRIGGER	RULE
Purchase Value > ₱5,000,000	Always negotiate price
Delivery > 30 Days	Negotiate delivery
Warranty < Company Standard	Negotiate warranty

Communication composer

- Email · SMS · Supplier Portal
 - Templates · AI Draft · Manual Edit
 - Attachment Manager
 - Preview Before Sending
-

AI draft generator

Example: "Based on your quotation, we appreciate your proposal. To move forward, we request consideration of the following revisions..." The buyer can edit before sending. Nexi never sends communications automatically unless company policy explicitly enables it.

Internal collaboration

Purchasing can request input from Engineering · Production · Maintenance · Accounting · Management. Internal discussions remain invisible to suppliers.

Negotiation dashboard

- Average Savings
- Negotiation Duration

- Suppliers Active
- Pending Responses
- Average Response Time
- Negotiation Success Rate

Security

Only assigned buyers can negotiate. Managers can monitor. Executives have read-only access. All negotiations are encrypted. Digital signatures optional.

APIs

METHOD + PATH	PURPOSE
GET /api/negotiation/{rfqId}	Get negotiation state
POST /api/negotiation/message	Send message
POST /api/negotiation/email	Email dispatch
POST /api/negotiation/sms	SMS dispatch
POST /api/negotiation/revision	Record offer revision
GET /api/negotiation/history	Full history
POST /api/ai/negotiation-advice	Nexi advice packet

Database tables

- Negotiation
- NegotiationMessages
- NegotiationOffers
- NegotiationRevisions
- NegotiationAttachments
- NegotiationConcessions

- InternalComments
- AuditLog

KPIs

- Average Cost Reduction
- Average Negotiation Time
- Negotiation Success Rate
- Average Warranty Improvement
- Average Delivery Improvement
- Supplier Acceptance Rate
- Savings Generated

Acceptance criteria

- Every negotiation message is permanently linked to the RFQ and supplier.
- All supplier offer revisions are version-controlled.
- AI suggestions are advisory and require user confirmation before communication.
- Concessions are tracked by category and visible in the negotiation history.
- Internal discussions remain isolated from supplier communications.
- Complete audit history is retained for every negotiation action.

ENHANCEMENT – NEGOTIATION INTELLIGENCE MEMORY

Nexi should learn from every completed negotiation. For each supplier she builds a negotiation profile: typical discount range accepted · avg number of negotiation rounds · preferred channel · avg response time · areas of flexibility · seasonal pricing trends · common reasons for rejecting requests. New negotiations start with strategies tailored to that specific supplier.

SCREEN 008 · ACC-PUR-008

AI Award Recommendation

Nexi recommends. An authorized approver decides.

Purpose

After negotiations are complete, Nexi evaluates all final offers and recommends the supplier(s) that provide the best overall value.

IMPORTANT RULE

Nexi NEVER awards a contract. Nexi only recommends. An authorized company approver makes the final decision.

Entry conditions

- Negotiation completed
- Final quotations received
- Technical evaluation completed
- Commercial evaluation completed
- Budget approved

Layout — three columns

Left panel — procurement summary

- Purchase Request
- RFQ

- Buyer
- Department
- Project
- Budget
- Currency
- Total Purchase Value
- Number of Suppliers Evaluated
- Negotiation Completed

Supplier ranking

RANK	SUPPLIER	SCORE
1	ABC Industrial	96.8
2	XYZ Engineering	94.1
3	Prime Industrial	90.5

Center panel — award comparison matrix

Each criterion displays Score · Weight · Weighted Score · Reason. Criteria:

- Commercial
- Technical
- Delivery
- Warranty
- Risk
- Financial
- Support
- Past Performance
- Sustainability
- Compliance
- Innovation
- Contract Terms

Total Cost of Ownership

- Purchase Price
- Freight

- Insurance
- Taxes
- Import Duties
- Installation
- Commissioning
- Training
- Maintenance
- Expected Spare Parts
- Five-Year Ownership Cost

Not just purchase price.

Value analysis

	SUPPLIER A	SUPPLIER B	RECOMMENDATION
Purchase Price	₱4,800,000	₱4,950,000	Supplier B
Five-Year Cost	₱5,300,000	₱5,020,000	Lower TCO

Right panel — Nexi recommendation

- Recommended Supplier
- Confidence
- Overall Score
- Business Risk
- Savings
- Reasoning
- Expected Delivery
- Expected ROI
- Decision Summary

Explainable AI

Every recommendation includes: Why this supplier · Why not the others · Risks · Trade-offs · Financial impact · Operational impact · Confidence level. Nothing is hidden.

Recommendation modes

- Lowest Price
- Best Value
- Fastest Delivery
- Lowest Risk
- Highest Quality
- Lowest Total Cost of Ownership
- Custom Company Policy

Purchasing Management configures the default.

Multi-award

Supports Single Supplier · Split Award · Regional Award · Category Award · Percentage Award (e.g. Supplier A 70% · Supplier B 30%).

Scenario simulator

Buyer can test: "What if Supplier A reduces price by 5%?" · "What if Supplier B delivers in 10 days?" · "What if warranty becomes 5 years?" Every scenario recalculates the ranking instantly. Simulations do not modify live data.

Conflict detection

- Supplier is suspended
- Supplier exceeds concentration policy
- Budget exceeded
- Contract conflict
- Missing approvals
- Compliance expired
- Insurance expired

- Required certifications missing

Executive summary (one page)

- Purchase Objective
- Recommended Supplier
- Alternative Supplier
- Financial Summary
- Key Risks
- Key Benefits
- Savings
- Approval Recommendation
- Export to PDF

Decision options

- Approve Recommendation
- Select Different Supplier
- Split Award
- Return for Negotiation
- Reject All Quotations
- Cancel Procurement
- Escalate to Executive Review

Override rules

If the approver selects a supplier other than Nexi's recommendation, mandatory fields: Reason · Business justification · Supporting documents (optional) · Second approval (if required). Override is permanently recorded.

Approval preview

Purchasing Manager → Accounting Manager → Finance Director (by value) → CEO (by approval limit) → Board (optional, configurable)

Security

Only authorized approvers can finalize recommendations. Nexi cannot approve purchases. Recommendations are immutable after submission; any change creates a new revision with a full audit trail.

APIs

METHOD + PATH	PURPOSE
GET /api/award/recommendation/{rfqId}	Award recommendation packet
POST /api/award/submit	Submit award decision
POST /api/award/override	Record override + justification
GET /api/award/scenarios	Load / save what-if scenarios
GET /api/approval/matrix	Live approval matrix
POST /api/report/executive-summary	Generate one-page executive summary

Database tables

- AwardRecommendation
- AwardScenario
- AwardApproval
- AwardOverride
- SupplierRanking

- CostOwnership
- ApprovalWorkflow
- AuditLog

KPIs

- AI Recommendation Acceptance Rate
- Average Procurement Savings
- Total Cost of Ownership Savings
- Award Cycle Time
- Override Rate
- Supplier Concentration Index
- Procurement Risk Score

Acceptance criteria

- Supplier rankings are calculated from configurable weighted criteria.
- Total Cost of Ownership is included in every recommendation.
- Scenario simulations do not modify live procurement data.
- Any override requires a reason and is permanently audited.
- The selected supplier automatically advances to Purchase Order generation after all required approvals.
- A one-page executive summary can be generated for every award decision.

ENHANCEMENT – CORPORATE PROCUREMENT MEMORY

Beyond a single procurement, Nexi should remember how previous award decisions performed. For every awarded supplier she tracks: Did they deliver on time? Did final cost match the quotation? Were there quality issues? How many warranty claims? How many change orders? Was the buyer satisfied? Over time she recommends based on actual business outcomes, not just on quotations.

SCREEN 009 · ACC-PUR-009

Purchase Order Generation

Official company obligation — extreme control required

Purpose

Generate the official Purchase Order (PO) from the approved supplier award. The PO is generated automatically from approved data. Users should never retype information that already exists in the PR, RFQ, quotation, or award recommendation.

Entry conditions

- Award approved
 - Budget approved
 - Approval workflow completed
 - Supplier selected
 - No blocking compliance issues
-

Purchase Order number

PO-YYYY-NNNNNN (example: PO-2026-001254)

Sequential · company configurable · immutable after creation.

Header

- PO Number
- Supplier
- Purchase Request
- RFQ
- Award Reference
- Buyer
- Department
- Project
- Cost Center
- Currency
- Issue Date
- Required Delivery Date
- PO Status

Status workflow

Draft → Pending Approval (if amendment) → Issued → Supplier Acknowledged → Partially Delivered → Fully Delivered → Closed → Archived

Supplier information (auto-filled)

- Company Name
- Address
- Contact Person
- Email
- Telephone
- Tax ID
- Payment Terms
- Delivery Terms

- Incoterms
- Bank Details (view restricted)

Item grid

- Line Number
- Item Code
- Description
- Technical Specification Reference
- Quantity
- Unit
- Unit Price
- Discount
- Tax
- Freight Allocation
- Line Total
- Delivery Date
- Delivery Location
- Warehouse
- Remarks

Every field is inherited from approved procurement documents unless explicitly editable.

Financial summary

- Subtotal
- Discount
- Tax
- Shipping
- Insurance
- Other Charges
- Grand Total
- Budget Remaining
- Commitment Amount

- Accounting GL Preview

Attachments (auto-attached)

- Approved Purchase Request
- RFQ
- Final Supplier Quotation
- Award Recommendation
- Technical Drawings
- Specifications
- Contract (if applicable)

No manual searching required.

Purchase terms

- Payment Terms
- Delivery Terms
- Warranty
- Penalty Clauses
- Quality Requirements
- Inspection Requirements
- Packing Requirements
- Required Documentation
- Insurance Requirements
- Acceptance Criteria

Delivery schedule

Supports Single · Partial · Milestone · Scheduled Delivery. Each delivery has: Date · Quantity · Destination
· Warehouse · Responsible Receiver.

Compliance verification (before issue)

- Supplier active
- Required approvals completed
- Budget still available
- Certifications valid
- Insurance valid
- Contract valid
- Sanctions / blacklist checks (if enabled)
- Required attachments present

If any mandatory rule fails, PO issuance is blocked.

Supplier confirmation

Supplier receives the PO via Email · Supplier Portal · EDI/API (optional). Supplier options: Accept · Reject · Request Amendment · Propose Delivery Change. Every response tracked.

PO amendments

A new revision is created. Original PO remains immutable. Every amendment records: revision number · date · user · reason · approval (if required) · supplier acknowledgment.

Integration (after issue)

- Commit budget in Accounting
- Create PO in SOS Inventory
- Notify Warehouse
- Notify Requester
- Notify Finance (if configured)
- Start delivery tracking

User actions

- Generate PO
- Preview
- Validate
- Issue
- Print PDF
- Email Supplier
- Export PDF
- Export Excel
- Create Amendment
- Cancel (before issue only)

Security

Only authorized Purchasing users can issue a PO. Only authorized approvers can approve amendments. Original PO cannot be edited after issue. Digital signatures supported. Every action written to the audit log.

APIs

METHOD + PATH	PURPOSE
POST /api/purchase-orders/create	Create PO from approved award
POST /api/purchase-orders/issue	Issue PO (validate + commit)
POST /api/purchase-orders/amend	Create amendment
GET /api/purchase-orders/{id}	Fetch PO
POST /api/purchase-orders/send	Dispatch to supplier
POST /api/sos/purchase-order	Push PO into SOS
POST /api/accounting/commit-budget	Reserve budget commitment

Database tables

- PurchaseOrder
- PurchaseOrderItems
- PurchaseOrderRevisions
- PurchaseOrderAttachments
- PurchaseOrderTerms
- PurchaseOrderDeliverySchedule
- PurchaseOrderAcknowledgement
- BudgetCommitments
- AuditLog

Notifications

- On PO issue: Supplier · Requester · Purchasing Manager · Warehouse · Accounting · Finance (optional)
- On supplier acknowledgment: Assigned Buyer · Purchasing Manager
- On amendment request: Assigned Buyer · Purchasing Manager · Approvers (if required)

KPIs

- PO Cycle Time
- Average Time from Award to PO
- Supplier Acknowledgment Time
- Amendment Rate
- Budget Commitment Accuracy
- On-Time PO Issuance

Acceptance criteria

- PO data is generated from approved records without manual re-entry.

- Mandatory validations are completed before issue.
- Issued POs are immutable; changes require controlled amendments.
- Supplier acknowledgments are tracked and linked to the PO.
- Budget commitments and SOS integration occur automatically upon issue.
- All actions and revisions are fully auditable.

ENHANCEMENT – INTELLIGENT PURCHASE ORDER

The PO becomes a live operational object that continuously monitors: supplier acknowledgment status · delivery milestones · shipment delays · budget impact · contract compliance · upcoming receiving activities. If a supplier misses a milestone or requests a delivery change, Nexi immediately alerts Purchasing and updates the operational timeline.

SCREEN 010 · ACC-PUR-010

Goods Receiving Center (GRN)

Supplier delivery → warehouse acceptance → quality inspection → inventory posting

Purpose

Manage the complete receiving process from supplier delivery through warehouse acceptance, quality inspection, inventory posting, and financial matching.

IMPORTANT PRINCIPLE

Receiving does NOT automatically mean acceptance. Goods can be: Accepted · Partially Accepted · Rejected · Quarantined · Returned.

Entry conditions

- Purchase Order Issued
- Supplier Delivery Scheduled
- Supplier Shipment Notice Received (Optional)

Layout — three panels

Left panel — incoming deliveries

- Expected Today
- Expected Tomorrow
- Overdue Deliveries
- Deliveries Waiting Inspection

- Partial Deliveries
- Urgent Deliveries

Selecting one opens the receiving workspace.

Center panel — delivery details header

- Receiving Number (GRN-YYYY-NNNNNN)
- Purchase Order
- Supplier
- Truck Number
- Driver
- Receiving Warehouse
- Receiving Date
- Receiver
- Delivery Status

Item receiving grid

- Item Code
- Description
- Ordered Qty
- Delivered Qty
- Accepted Qty
- Rejected Qty
- Remaining Qty
- Unit
- Batch Number
- Serial Number
- Expiry Date (if applicable)
- Warehouse Location
- Inspection Status
- Remarks

Barcode / QR / RFID

Each line supports Barcode Scan · QR Scan · RFID Scan · Manual Entry (permission controlled). After scanning, system validates: correct item · correct PO · correct quantity · duplicate scan · wrong warehouse.

Photo documentation

Receiver may attach: photos · videos · documents · damage reports · packing lists · delivery receipts. Every file becomes part of the permanent receiving record.

Quantity validation

FIELD	EXAMPLE
Ordered	100
Received Previously	30
Delivered Today	50
Remaining	20

Over-delivery / under-delivery

- Over-delivery: system displays Over Delivery flag, Approval Required, configurable company policy (Accept · Reject · Partial Accept)
- Under-delivery: system calculates Outstanding Balance · Expected Date · Back Order · Supplier Commitment

Quality inspection

Pending Inspection → Passed → Accepted Failed → Quarantine → Return to Supplier

Inspection checklist

- Visual Inspection
- Packaging
- Label Verification
- Specification Match
- Damage
- Dimensions
- Certificates
- Temperature
- Batch Verification
- Expiry Verification
- Custom Checklist

Warehouse posting

Only after acceptance. Automatically updates: SOS Inventory · Warehouse Stock · Available Stock · Reserved Stock · Inventory Valuation · Lot Tracking · Serial Numbers.

Right panel — Nexi receiving assistant

- Expected Quantity
- Supplier History
- Past Defects
- Recommended Inspection Level
- Risk Score
- Delivery Performance
- Warranty Alerts

- Special Instructions

AI receiving analysis

- Frequent Supplier Short Deliveries
- Repeated Damages
- Wrong Packaging
- Repeated Batch Problems
- Late Deliveries
- Temperature Deviations
- Unexpected Serial Numbers
- Repeated Quality Issues

Exceptions

- Damage
- Missing Items
- Wrong Items
- Incorrect Quantity
- Expired Products
- Failed Inspection
- Broken Packaging
- Missing Documentation
- Wrong Batch
- Wrong Serial Number

Every exception creates an incident record.

Return to Supplier (RTS)

If rejected, automatically prepares: Return Authorization · Supplier Notification · Return Items · Reason · Photos · Replacement Request · Credit Request · Replacement Deadline.

Financial matching — three-way match

Purchase Order → Goods Received → Supplier Invoice

Only matching transactions proceed to payment unless an authorized exception is approved.

User actions

- Receive
- Scan
- Accept
- Reject
- Partial Accept
- Quarantine
- Return
- Transfer
- Print GRN
- Generate Incident
- Upload Photos

Security

Warehouse cannot modify PO prices. Accounting cannot modify received quantities. Quality can only change inspection status. Only Purchasing can authorize supplier-related exceptions. Every action requires authentication and is audited.

APIs

METHOD + PATH	PURPOSE
POST /api/grn/create	Create GRN header
POST /api/grn/receive	Record received quantities
POST /api/grn/inspection	Record inspection results
POST /api/grn/accept	Accept quantities into stock
POST /api/grn/reject	Reject quantities
POST /api/grn/return	Create Return to Supplier
POST /api/sos/inventory/receive	Push accepted stock into SOS
GET /api/po/{id}	PO context
GET /api/supplier/history	Supplier history for assistant panel
POST /api/photos/upload	Photo / evidence upload

Database tables

- GoodsReceipt
- GoodsReceiptItems
- InspectionResults
- InspectionChecklist
- Returns
- WarehouseTransactions
- InventoryLots
- InventorySerials
- ReceivingPhotos
- ReceivingIncidents
- AuditLog

Notifications

- Receiving Completed
- Inspection Failed
- Return Created
- Supplier Replacement Required
- Warehouse Updated
- Accounting Ready for Invoice Matching
- Purchasing Exception Required

KPIs

- Receiving Accuracy
- Average Receiving Time
- Inspection Pass Rate
- Supplier Defect Rate
- Supplier Short Delivery Rate
- Average Time to Stock Availability
- Return Rate
- Three-Way Match Success Rate

Acceptance criteria

- Goods cannot be received against an invalid or closed PO.
- Quantity validations prevent unauthorized over-receiving unless approved by policy.
- Inventory is updated only after accepted quantities are confirmed.
- Inspection results determine inventory availability (Accepted vs. Quarantine).
- All receiving exceptions generate traceable incident records.
- Every receiving transaction is linked to the originating PO and fully audited.

ENHANCEMENT – INTELLIGENT RECEIVING

When a shipment arrives, Nexi analyses: supplier defect rate · delivery punctuality · shipment value · product criticality · regulatory requirements · past inspection failures. She dynamically recommends the inspection level: L1 Visual · L2 Random sampling · L3 Full quality inspection · L4 Mandatory laboratory / engineering verification before inventory release.

SCREEN 011 · ACC-PUR-011

Invoice Verification & Three-Way Matching

Protect the company from overbilling, duplicates, fraud

Purpose

Verify supplier invoices against Purchase Order · Goods Receipt (GRN) · Supplier Invoice. No invoice proceeds to payment unless company rules are satisfied.

Protects the company from: overbilling · duplicate invoices · incorrect quantities · incorrect prices · fraud · payment mistakes.

Entry conditions

- Goods Received
- Supplier Invoice Received
- Purchase Order Exists

Header

- Invoice Number
- Supplier
- PO Number
- GRN Number
- Invoice Date

- Due Date
- Currency
- Invoice Status
- Assigned Accountant
- Assigned Buyer

Status workflow

Invoice Received → OCR Processing → AI Validation → Three-Way Match → Exception Review (if required) → Approved → Payment Queue → Paid → Closed

Left panel — invoice information

- Invoice Amount
- Supplier
- Purchase Order
- Delivery
- Tax
- Payment Terms
- Invoice Currency
- Exchange Rate
- Attachments

Invoice attachments

- PDF
- XML
- Images
- Scanned Documents
- Government e-Invoice (future)

Center panel — three-way matching

FIELD	PO	GRN	INVOICE	STATUS
Item	✓	✓	✓	Match
Quantity	100	100	100	Match
Unit Price	₱120	₱120	₱120	Match
Tax	12%	12%	12%	Match
Freight	₱2,000	₱2,000	₱2,000	Match

Every discrepancy is highlighted.

Matching rules

Exact Match · Tolerance Match · Manual Review. Company configurable. Example — Quantity tolerance $\pm 2\%$ · Price tolerance $\pm 1\%$ · Tax 0%.

AI invoice reading

Automatically extracts: Supplier · Invoice Number · Invoice Date · Items · Prices · Taxes · Discounts · Payment Terms · Bank Details · Reference Numbers. No manual encoding.

Duplicate invoice detection

Checks: Invoice Number · Supplier · Amount · Date · PO · GRN · Historical Payments. If duplicate suspected, payment blocked.

Right panel — Nexi financial assistant

- Invoice Risk
- Payment Risk
- Budget Impact
- Duplicate Probability
- Recommended Action
- Confidence

Example — Invoice matches 99.7% · Duplicate Risk 0% · Recommendation: Approve.

Exceptions

- Price Difference
- Quantity Difference
- Tax Difference
- Freight Difference
- Missing GRN
- Missing PO
- Expired PO
- Closed PO
- Incorrect Currency
- Wrong Supplier

Exception resolution

Assign to Purchasing · Accounting · Warehouse · Supplier · Engineering · Legal. Each exception receives Owner · Priority · Deadline · Status.

Payment readiness

- Approved
- Waiting Approval
- Blocked

- Pending Exception
- Ready for Payment

Invoice timeline

Invoice Received → OCR Complete → Three-Way Match → Approved → Payment Scheduled → Paid

Payment information

- Payment Terms
- Due Date
- Discount Date
- Early Payment Discount
- Penalty Date
- Bank Account
- Payment Method

Fraud detection

- Duplicate invoices
- Changed bank accounts
- Repeated invoice numbers
- Abnormal invoice values
- Invoice splitting
- Suspicious tax
- Repeated supplier corrections
- Invoice timing anomalies

Buyer / accountant actions

- Approve
- Reject
- Request Supplier Correction
- Escalate
- Add Comment
- Assign Exception
- Export Report
- Print

Security

Only Accounting approves invoices. Purchasing can review commercial discrepancies. Warehouse cannot approve invoices. Bank account changes require dual approval. Every financial change is audited.

APIs

METHOD + PATH	PURPOSE
POST /api/invoice/upload	Upload invoice file
POST /api/invoice/ocr	OCR + extract
GET /api/invoice/match	Three-way match result
POST /api/invoice/approve	Approve invoice
POST /api/invoice/reject	Reject invoice
GET /api/payment/status	Payment queue status
POST /api/fraud/check	Run fraud checks

Database tables

- Invoices
- InvoiceItems
- InvoiceMatching
- InvoiceExceptions
- InvoiceApprovals
- PaymentSchedule
- SupplierBankAccounts
- FraudAlerts
- AuditLog

Notifications

- Invoice Received
- Matching Failed
- Invoice Approved
- Payment Ready
- Duplicate Invoice Alert
- Supplier Correction Requested

KPIs

- Invoice Match Rate
- Average Invoice Approval Time
- Duplicate Invoice Prevention
- Early Payment Discount Captured
- Invoice Exception Rate
- Average Payment Cycle
- Fraud Alerts Generated

Acceptance criteria

- Every invoice is automatically matched against the PO and GRN.
- OCR extracts invoice data with confidence scoring; low-confidence fields flagged for review.
- Duplicate invoice detection blocks payment until resolved.
- Tolerance rules are configurable by company policy.
- Invoice approval follows role-based permissions.
- All financial decisions and exceptions are fully audited.
- Approved invoices move automatically to the Accounts Payable payment process.

ENHANCEMENT — AUTONOMOUS FINANCIAL VALIDATION

Nexi calculates a Financial Confidence Score (0-100) per invoice — PO Match · GRN Match · Supplier History · Pricing Consistency · Duplicate Risk · Bank Account Verified · Invoice Pattern Normal. Policy can then define: 95-100 auto-route for Accounting approval · 80-94 require Purchasing review · <80 block payment and create investigation task. Human decides — Nexi assists.

SCREEN 012 · ACC-PUR-012

Supplier Performance & Scorecard

Suppliers become continuously evaluated business partners

Purpose

This screen transforms suppliers from static records into continuously evaluated business partners. Every PO, delivery, GRN, quality inspection, invoice, negotiation and contract automatically updates the supplier's performance score.

The score is system-generated. Users may add comments but cannot manually inflate supplier ratings.

Objectives

- Provide management with objective supplier performance
 - Support future purchasing decisions
 - Identify preferred suppliers
 - Detect supplier deterioration before it becomes a business problem
-

Entry conditions

Available after at least one completed Purchase Order.

Layout — three-column dashboard

Left panel — supplier directory

- Supplier Name
- Overall Score
- Status
- Preferred Supplier
- Conditional Supplier
- Suspended
- Blacklisted

Search / filter by: Category · Country · Risk · Buyer · Product Category.

Center panel — Supplier 360°

Overall Score 0-100 displayed as a circular gauge. Example: 96.4.

Performance breakdown

- Quality
- Delivery
- Price Stability
- Commercial Performance
- Communication
- Warranty Support
- Technical Support
- Invoice Accuracy
- Contract Compliance
- Sustainability

Each receives an individual score.

Trend

Last 30 days · 90 days · 12 months · 3 years. Shows improving or declining performance.

Right panel — Nexi supplier intelligence

- Supplier Risk
 - Expected Performance
 - Future Capacity
 - Recommended Action
 - Preferred Supplier · Conditional Supplier · Suspend Supplier · Increase Business · Reduce Business · Review Required
-

Automatic KPI calculation

Delivery

- On-Time Delivery %
- Average Delay
- Average Early Delivery
- Emergency Deliveries
- Missed Deliveries

Quality

- Accepted Deliveries
- Rejected Deliveries
- Inspection Failures
- Defect Rate
- Replacement Time
- Warranty Claims
- Corrective Actions

Commercial

- Average Price
- Price Stability
- Negotiation Success
- Discount History
- Cost Savings Generated
- Contract Compliance

Communication

- Average Response Time
- Average RFQ Response
- Negotiation Response
- Issue Resolution Time

Financial

- Invoice Accuracy
- Credit Note Accuracy
- Payment Disputes
- Bank Change Requests
- Financial Risk

Sustainability (configurable)

- Environmental Certifications
- Ethical Compliance
- Carbon Initiatives
- Waste Management

AI risk prediction

- Likelihood of Late Delivery
- Likelihood of Quality Failure
- Financial Instability
- Supplier Capacity Constraints
- Price Increase Probability
- Supply Chain Risk
- Supplier Dependency

Risk levels: Low · Medium · High · Critical.

Performance timeline

Chronological history of: POs · Deliveries · Returns · Quality Incidents · Negotiations · Contracts · Invoices · Corrective Actions · Audits. Every event clickable.

Supplier Improvement Plan

When a supplier falls below a configurable threshold, Nexi creates an improvement plan.

FIELD	VALUE
Issue	Late Deliveries
Target	95% On-Time
Review Date	30 Days
Owner	Purchasing Manager / Supplier Representative
Progress	Automatically tracked

Preferred Supplier logic (example thresholds)

- Overall Score ≥ 90
- Quality ≥ 95
- Delivery ≥ 95
- Risk = Low
- No major non-conformance in last 12 months → Preferred Supplier

If the supplier falls below policy thresholds, Nexi automatically recommends a review.

Blacklist management

Only authorized management may suspend or blacklist suppliers. Reasons include: repeated quality failures · fraud · compliance violations · contract breaches · safety violations · legal issues. Every blacklist action requires approval and justification.

Buyer / Manager actions

- View Supplier
- Compare Suppliers
- View Purchase History
- View Contracts
- Start Corrective Action
- Schedule Supplier Audit
- Create Improvement Plan
- Recommend Preferred Supplier
- Recommend Suspension
- Export Scorecard

Security

Performance metrics are system-generated. Manual changes not permitted. Management actions require role-based approval. Every action audited.

APIs

METHOD + PATH	PURPOSE
GET /api/suppliers/scorecard	Overall + breakdown scores
GET /api/suppliers/performance	KPI panel
GET /api/suppliers/timeline	Chronological history
GET /api/suppliers/risk	Risk prediction
POST /api/suppliers/improvement-plan	Create improvement plan
POST /api/suppliers/status	Change status (with approval)

Database tables

- SupplierPerformance
- SupplierKPIs
- SupplierRisk
- SupplierImprovementPlans
- SupplierStatusHistory
- SupplierAudits
- CorrectiveActions
- SupplierTimeline
- AuditLog

KPIs

- Overall Supplier Score
- On-Time Delivery
- Quality Acceptance Rate
- Average Negotiation Savings
- Average Response Time
- Invoice Accuracy
- Warranty Claim Rate
- Supplier Risk Index
- Preferred Supplier Ratio
- Supplier Improvement Success Rate

Acceptance criteria

- Every completed procurement transaction automatically updates supplier KPIs.
- Overall supplier scores are calculated from configurable weighting rules.
- Historical trends are retained for long-term analysis.
- Improvement plans can be tracked through completion.
- Preferred and suspended supplier status changes require authorization and are fully audited.

- Future purchasing decisions can reference the current supplier score and risk level without manual intervention.

ENHANCEMENT – SUPPLIER CAPACITY INTELLIGENCE

Before issuing a large RFQ or PO, Nexi estimates whether the supplier can realistically fulfill the demand by considering: historical fulfillment rates · current open POs · seasonal demand patterns · manufacturing lead times · production capacity (where available) · existing commitments to HydroNexis. Answers "Can this supplier successfully deliver this order now?"

SCREEN 013 · ACC-PUR-013

Procurement Analytics & Executive Intelligence

Executive Decision Center — read-only for most users

Purpose

Provide executives with complete visibility into procurement performance, costs, supplier health, risks, savings, and future purchasing requirements. This screen is read-only for most users. No operational work is performed here. It is an Executive Decision Center.

Users

ROLE	ACCESS
CEO	Full
President	Full
CFO	Full
Accounting Manager	Full
Purchasing Manager	Full
Department Managers	Limited
Buyers	Personal KPIs only

Dashboard layout

Four-row executive dashboard. No scrolling on a 32-inch monitor. Every widget drill-down capable.

Executive KPI row — Procurement spend

- Today
- This Month
- This Year
- Budget
- Forecast

Savings

- Negotiation Savings
- Contract Savings
- Inventory Transfer Savings
- AI Savings
- Emergency Purchase Avoidance

Purchase cycle

- Average PR Approval Time
- Average RFQ Time
- Average Negotiation Time
- Average PO Generation
- Average Receiving Time
- Average Invoice Approval
- End-to-End Procurement Cycle

Procurement Health Score

Overall Score 0-100 calculated from Supplier Performance · Budget Compliance · Cycle Time · Risk · Savings · Quality.

Spend analytics

Interactive charts. Spend by: Supplier · Category · Department · Project · Warehouse · Region · Buyer · Month · Quarter · Year. Every chart supports drill-down.

Budget intelligence

- Approved Budget
 - Committed Budget
 - Actual Spend
 - Forecast Spend
 - Remaining Budget
 - Over Budget Projects
 - Department Utilization
 - Cash Flow Forecast
-

Supplier intelligence

- Top Suppliers
 - Highest Risk Suppliers
 - Most Improved Suppliers
 - Suppliers Needing Review
 - Preferred Suppliers
 - Blacklisted Suppliers
 - Supplier Dependency
 - Supplier Concentration
-

Procurement risk map

Heat map. Categories: Supplier Risk · Budget Risk · Delivery Risk · Contract Risk · Compliance Risk · Currency Risk · Single Source Risk · Inventory Risk. Each area clickable.

AI insights (examples)

- Purchasing costs increased 8% compared with last month.
- Supplier ABC has delivered late five consecutive times.
- Annual contract renewal due in 21 days.
- Combining three purchase requests will save approximately ₱425,000.
- Switching to Supplier XYZ could reduce annual cost by 12%.

Every insight links directly to supporting transactions.

Procurement forecast

- Future Spend
- Supplier Capacity
- Budget Exhaustion
- Contract Expiration
- Inventory Shortages
- Price Trends
- Demand Peaks
- Cash Requirements

Forecast periods: 30 Days · 90 Days · 12 Months · 3 Years.

Executive alerts (critical only)

- Budget Exceeded
- Supplier Bankruptcy Risk
- Repeated Quality Failures
- Contract Expiration
- Late Deliveries
- Single Supplier Dependency
- Compliance Failure
- Fraud Investigation
- Approval Bottlenecks

Executive drill down

Every KPI opens Supplier · Purchase Order · Project · Buyer · Department · Invoice · Contract · Timeline.
No separate search required.

Benchmarking

- This Month vs Last Month
 - This Quarter vs Previous Quarter
 - This Year vs Previous Year
 - Company vs Target
 - Department vs Department
 - Buyer vs Buyer
-

AI executive recommendations — Top 10 opportunities

- Reduce Spend
- Improve Supplier Mix
- Renegotiate Contracts
- Increase Inventory Transfers
- Reduce Emergency Purchases
- Consolidate Suppliers
- Improve Payment Terms
- Extend Contracts
- Increase Competition
- Reduce Procurement Cycle

Every recommendation includes estimated annual financial impact.

Export

- PDF

- Excel
- PowerPoint
- Executive Board Pack
- Monthly Procurement Report
- Quarterly Report
- Annual Report

Security

Executives view all data. Managers view assigned departments. Buyers view only their procurement activities. Financial values follow role-based permissions. Every report export is logged.

APIs

METHOD + PATH	PURPOSE
GET /api/analytics/procurement	Overall procurement KPIs
GET /api/analytics/spend	Spend by dimension
GET /api/analytics/suppliers	Supplier analytics
GET /api/analytics/budget	Budget analytics
GET /api/analytics/risk	Risk heat map
GET /api/analytics/forecast	Forecast payloads
GET /api/analytics/insights	AI insight feed
POST /api/report/export	Export report (PDF / Excel / PPTX)

Database tables

- AnalyticsCache
- ProcurementKPIs
- ExecutiveInsights

- ForecastData
- RiskDashboard
- SpendAnalytics
- SupplierAnalytics
- BudgetAnalytics
- AuditLog

KPIs

- Total Spend
- Annual Savings
- Negotiation Savings
- Average Procurement Cycle
- Budget Compliance
- Supplier Performance Index
- Contract Utilization
- Procurement ROI
- Emergency Purchase Rate
- AI Recommendation Adoption Rate

Acceptance criteria

- Dashboard loads in under 5 seconds using pre-aggregated analytics where appropriate.
- Every KPI supports drill-down to the underlying transactions.
- Forecasts are recalculated on configurable schedules.
- AI insights are traceable to supporting procurement data.
- Executive exports reproduce the dashboard accurately.
- All financial data respects role-based security and audit requirements.

ENHANCEMENT – EXECUTIVE PROCUREMENT DIGITAL TWIN

Nexi creates a live simulation of the procurement operation. Executives test scenarios like "What if Supplier A raises prices by 15%?" · "What if we consolidate into three annual contracts?" · "What if we reduce emergency purchasing by 50%?" · "What if we shift 20% of spend to local suppliers?" · "What if demand rises 30% next quarter?" The system immediately projects impacts on cash flow · budget · procurement cycle time · supplier workload · inventory · operational risk · estimated annual savings.